EventFrame Whitepaper

A Mathematical Framework for Event-Centric Prediction

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Public working paper. This version has not been experimentally validated.

Abstract

EventFrame is a framework for event-centric prediction. It represents experience as typed event frames rather than as unstructured sequences alone, but does not treat those frames as fundamental ontology. Event frames are task-relative compressed records. For physical substrates, physical information bounds motivate a limiting thought experiment about microscopic description; they do not prove a discrete substrate or the framework's sparsity hypothesis, and they do not support claims about simulated or software substrates.

The central object is $e_t \in \mathcal{E}_{\Delta_\tau}$, obtained as $e_t = \Gamma_{\Delta_\tau}(\omega_{A_t})$. Temporal resolution may range from seconds to microseconds when measurement supports it. Given $C_t = e_{t-k+1:t}$, the predictor returns a distribution over marked event times and a no-event outcome. A strictly proper score is the primary probabilistic-fidelity metric; the governing system-design objective is composite, and bounded event-aware timing error is diagnostic.

At a fixed temporal resolution, a target-law constrained population objective supplies an oracle benchmark. The operational rule instead minimizes empirical priority-weighted post-observation action plus non-negative representation cost over a finite family satisfying certified abstraction and proper-score constraints. Oracle feasibility and empirical certifiability are distinct. Candidate design and untouched chronological confirmation are separated. Prediction-time admission uses a distinct risk and state containing only quantities available before the prediction occurs.

EventFrame separates a baseline from cached statistical error correction. A horizon-indexed residual record carries separately typed point-template and forecast-law components, each with its own clipping and estimator semantics. Proper-score evaluation requires the law component's declared Markov kernel on the complete measurable marked-time/no-event space; auxiliary structured fields use the point component. The kernel explicitly governs the no-event atom, and a fixed decision rule keeps the final mark and time coherent with the corrected law. A selective Bayesian frontier uses bounded vector retrieval, sheaf-inspired compatibility neighbors, and as-of causal or predictive adjacency to nominate belief updates. Activation controls update expenditure, while Anti-Pigeon certificates determine which events may share a posterior. Selection-conditioned likelihoods, independent inactive-event audits, changepoint invalidation, and explicit resource caps prevent the local posterior from being presented as an unqualified full-stream posterior. A separately typed operator composes runtime packets. The operator representation takes limited inspiration from Causal Fermion Systems; EventFrame's clipping, projection, and residual objective are its own constructions, not the CFS causal action.

Validity-constrained fuzzing measures model sensitivity, not causality. Causal claims require an explicit structural causal model and identified interventions. Approximate predictive lumpability compares detailed contexts that share an abstract context. Every group retains a traceability frame plus a coverage-aware context audit set; one representative alone cannot establish group stability. A staged compatibility layer can compare heterogeneous abstractions, reconcile local forecasts, perform bounded predictive sheaf snaps after untouched confirmation, apply spectral refinement under explicit linear assumptions, and preserve regime mixtures. Snaps revise predictive organization through atomic graph-key-epoch publication; they do not promote causal edges. Refinement depth is selected by priority, evidence, and measured hardware cost while certified cache

reuse remains available.

The paper remains a research framework without implementation results. A finite worked instantiation demonstrates the residual, no-event, horizon-gate, and operational-selection arithmetic but is not empirical validation. The evaluation program uses as-of rolling-origin replay and measures proper forecast quality, residual utility, Bayesian activation and selection calibration, omitted influence, changepoint detection, sensitivity stability, audit coverage, abstraction compatibility, predictive-snap accuracy and churn, priority-weighted correction, regime adaptation, and hardware-indexed runtime tradeoffs.

1. Introduction

Prediction systems often operate over sequences whose internal structure is only implicit. A model may receive tokens, vectors, logs, traces, or state observations and learn statistical regularities among them. This can be effective, but it makes some questions difficult to ask directly: which compressed distinction mattered, what changed, when did it happen, where did it occur, why might it matter, and how did it transform the state of the world?

EventFrame begins from a compression premise: a modeled substrate may contain more detail than a prediction system can retain. Event frames are task-relative coarse-grained representations, not assertions about fundamental spacetime. For physical substrates, Planck scales and information bounds motivate a limiting thought experiment [7–9]; they do not prove the compression premise. Simulated and software substrates require independent task- and resource-based justification.

The framework represents experience as event frames selected for predictive and intervention relevance. An event frame is a typed record of an occurrence or transition after compression. It includes the 5W1H fields of who, what, when, where, why, and how, plus auxiliary state and confidence metadata. The goal is not to claim that every domain naturally exposes these fields perfectly. The goal is to create a disciplined representation in which uncertainty, missing fields, competing explanations, and compression choices can still be recorded explicitly.

The core contribution is adaptive event abstraction. At fixed resolution, an external-law population objective defines the constrained oracle benchmark; the operational rule selects from a finite family using certified empirical constraints, post-observation action, and representation cost. It distinguishes model-sensitivity evidence from causal intervention evidence.

Given $C_t = e_{t-k+1:t}$, the system predicts a distribution over event identity, event time, and no event within horizon H. A proper forecast score is the primary probabilistic-fidelity metric because timing-only loss can reward the wrong event at the right time. The complete system design minimizes a separate composite objective under a proper-score guard. Event-aware timing remains an interpretable diagnostic.

The reference prediction procedure has seven steps:

1. Form a context C_t from the last k event frames.
2. Compute a baseline forecast law $\mathsf{Q}_B(\cdot \mid C_t)$ and conditional event template $b_t = B(C_t)$.
3. From candidate-specific state S_{Θ,t^-}, select an exact-key or general residual record $\mathbf{r}_t^{\text{use}}$ only when its distance, confidence, effective-support, age, epoch, forecast-horizon equality,

compatibility-margin, and component-provenance checks pass.

4. Construct a bounded Bayesian update frontier, apply the recorded activation rule, and share a cached posterior only under a current Anti-Pigeon certificate; otherwise preserve separate posteriors or defer.
5. Clip the separately typed point and law residual components, apply each only to its declared semantics, derive a mark/time-coherent no-event-capable summary from the corrected law, and apply the pre-observation risk gate to the complete output bundle.
6. Observe the next marked event or no-event outcome and evaluate proper predictive loss.
7. Use a slower refinement process to audit selective updates, detect changepoints, recalibrate beliefs, update residuals, test invariants, revise abstractions, or revise the event ontology.

This procedure explains why the framework includes both memory and residual prediction. Episodic memory stores prior cases. A residual cache stores reusable corrections to a baseline transition. The distinction matters because recalling a similar event and applying a similar error correction are not the same operation. The first supports case-based reasoning; the second supports low-latency approximation when similar contexts produce similar transition errors.

The slow path uses validity-constrained perturbations to test model sensitivity, coverage-aware bucket audits to find hidden divergence, and approximate predictive lumpability to test compression. Causal analysis is a separate optional path requiring structural equations and identification assumptions.

The contributions of this paper are therefore:

1. A compressed event-frame ontology for prediction-oriented representation.
2. A governing optimization principle for adaptive event abstraction.
3. A residual prediction model with constrained composition and action-residual fast-path caching.
4. A selective Bayesian update frontier with Anti-Pigeon posterior-sharing control, selection-aware semantics, changepoint invalidation, and independent audit sampling.
5. A combined episodic, residual, and bounded belief-memory architecture.
6. A validity-constrained sensitivity method for conditional invariants and ontology review.
7. A lumpability-based approach to abstraction.
8. A bounded predictive sheaf-snapping rule for validated local compatibility-graph revision.
9. A fast-path and slow-path reference runtime model.
10. Experiment designs for testing the framework's claims.

These are proposed as a research framework, not as validated results or a fixed implementation. Event-centric latent retrieval itself has prior art [10]. Streaming and sequential Bayesian methods provide prior work for incremental posterior approximation and changepoint monitoring [14–18], while Pattern Markov Chains and shift-aware sequential prediction provide narrower event-forecasting precedents [19,20]. D'Acunto, Di Lorenzo, and Barbarossa's *Networks of Causal Abstractions: A Sheaf-theoretic Framework* provides prior work on coordinating heterogeneous causal abstractions through network sheaves, restriction maps, connection Laplacians, global sections, and mixture causal models [13]. EventFrame's claimed contribution is the typed residual-error and evidence-controlled event-abstraction loop, including selective Bayesian activation, Anti-Pigeon posterior granularity, cache certificates, bounded predictive sheaf snapping, and priority- and hardware-aware staged integration. It does not claim to invent streaming Bayes, particle filtering, online changepoint detection, or event-pattern forecasting. The snapping term is EventFrame terminology for validated local predictive-structure revision, not a theorem or standard operation inherited from sheaf theory. The next section defines the event ontology.

Claims Register

This section states the paper's major claims as falsifiable targets. The claims are not treated as established results. Each one names what would need to be measured, proved, or falsified by later experiments.

Claim 0. At a fixed resolution, an external target-law constrained population objective defines an oracle benchmark. The operational rule selects from a finite family using certified Anti-Pigeon and proper-score constraints, empirical priority-weighted action, and representation cost. Oracle feasibility and empirical certifiability are distinct; generating laws are distinct from their realized design and untouched confirmation samples, and as-of runtime admission remains separate.

Claim 1. Structured event frames are useful predictive units if, for a declared task, they improve interpretability or temporal prediction relative to unstructured sequence records without hiding field-level error.

Claim 1a. Event frames are task-relative compressed representations rather than claims about fundamental ontology. Physical constants do not prove the compression hypothesis.

Claim 1b. Temporal precision controls frame granularity if changing the declared time resolution Δ_τ changes the candidate-frame set, cache pressure, and detectable divergence boundaries in measurable ways.

Claim 2. Residual caches reduce prediction cost or error when similar contexts or action signatures produce similar baseline errors and as-of, metadata-gated residual records improve forward-held-out loss often enough to justify lookup and maintenance. Point residuals are in-horizon only, while a separately declared law estimator covers marked and no-event outcomes. A typed cache record keeps point-template and law corrections semantically separate; the full kernel governs the no-event atom, and the final mark and time are coherent with the corrected law. Joint records require forward validation of the complete bundle.

Claim 2a. Runtime prediction packets are useful when a separately typed packet composition operator improves selection of memory nodes, graph edges, retrieval lane, compaction risk, response mode, or control branch on held-out packet loss.

Claim 2b. A selective Bayesian layer can preserve bounded local update cost when vector width, graph degree, activated frontier size, hypothesis dimension, and retained changepoint state are explicitly capped. Activation determines which evidence is updated; Anti-Pigeon certificates determine which evidence may share a posterior. Informative selection must enter the likelihood or the result is reported only as an activation-conditioned working posterior. Independent inactive-event audits and omitted-influence bounds test whether suppression has become overconfident.

Claim 3. Episodic memory and residual cache memory serve different roles because prior-case recall and prior-error correction can be independently useful or harmful under the same prediction context.

Claim 4. Validity-constrained property fuzzing exposes conditional model invariants and ontology review signals. It does not establish causal effects without an explicit SCM and identification strategy.

Claim 5. Approximate predictive lumpability provides a route to abstraction when projected event states preserve target-relevant transition behavior within a declared divergence threshold.

Claim 5a. Event streams can conjoin or diverge over time when multiple streams become prediction-equivalent under a merge threshold or when small distinctions amplify into target-distinct downstream futures.

Claim 5b. Each group retains at least one concrete traceability frame plus a coverage-aware context audit set; one representative alone is insufficient for group-level divergence claims, and unseen-context certification additionally requires exhaustive coverage or a verified continuity bound.

Claim 6. Anti-Pigeon rejects buckets whose externally evaluated context-conditional target-law diameter exceeds threshold. Model-forecast diameter is diagnostic only. Regime comparisons require common support or transport assumptions, and causal attribution requires separate intervention evidence.

Claim 6a. Predictively effective distinctions are sparse only when their held-out ablation ratio is small within a finite declared candidate set. A causal sparsity ratio is a separate quantity available only under identified interventions.

Claim 7. Fast-path and slow-path separation is computationally useful if low-latency prediction can reuse cached residuals while slower background work improves future predictions without blocking the current one.

Claim 7a. Expected exact-key lookup is history-independent only when context update, key construction, graph degree, key size, and cache size are bounded; fallback and maintenance costs remain explicit.

Claim 8. Heterogeneous abstractions can be tested through declared comparison maps and edge defects, complementing within-bucket Anti-Pigeon audits. The current construction is a sheaf-inspired compatibility scaffold; it becomes sheaf-theoretic only when the required map laws hold and causal only when SCM semantics are supplied.

Claim 8a. A full refinement architecture can retain certified residual reuse while adding compatibility audit, local reconciliation, bounded predictive sheaf snapping, spectral refinement under linear assumptions, and regime-mixture refinement. Hardware changes stage depth, not stage meaning.

Claim 8b. A predictive sheaf snap can improve a heterogeneous abstraction network when a finite local edit family is selected on chronological design data, fixed comparison obligations prevent deletion from masquerading as compatibility, the candidate is accepted only on later untouched evidence, and graph-key-epoch state is published atomically. Compatibility evidence alone does not promote causal edges.

Claim 8c. Upgrade value is evaluated by predeclared priority-weighted utility beside unweighted and stratified results; loss and resource percentages are not compared until converted to a common utility scale.

2. Event Ontology

EventFrame uses event frames as predictive units, not as fundamental ontology. The underlying substrate is assumed to contain more detail than the predictor can retain. That substrate may be physical, simulated, biological, robotic, or software-based. EventFrame treats a frame as a task-relative compressed representation. For physical substrates, Planck scales and physical information bounds motivate a limiting thought experiment about microscopic description; they do not prove a discrete substrate, a Planck-scale sampling lattice, or the EventFrame sparsity hypothesis. Simulated and software substrates require independent task- and resource-based compression arguments.

An event is therefore a structured representation of a change, occurrence, action, observation, or state transition after coarse-graining. An event frame records that compressed event in fields that can be compared, predicted, fuzzed, cached, and abstracted.

An event frame at index t is written:

$$e_t = (w_t, a_t, \tau_t, \ell_t, m_t, h_t, x_t, c_t)$$

where w_t denotes participating agents or entities, a_t denotes the action or occurrence type, τ_t denotes the time index or interval, ℓ_t denotes location or spatial context, m_t denotes motive, objective, causal explanation, or inferred driver, h_t denotes mechanism or process, x_t denotes auxiliary state, and c_t denotes confidence, provenance, or uncertainty metadata. When τ_t is an interval, the complete interval remains part of the frame; scalar timing formulas use a separately declared temporal anchor, defaulting to interval onset.

The conceptual role of this ontology is compression. It prevents prediction from treating history as a single undifferentiated sequence, but it also prevents prediction from pretending that every microscopic distinction deserves its own event identity. The fields ask different compressed questions. The ”what” field identifies an occurrence type. The ”when” field supports temporal prediction loss. The ”who” and ”where” fields localize the event. The ”why” and ”how” fields record explanatory hypotheses and mechanisms. The auxiliary state field allows symbolic, vector, graph, or latent variables to travel with the event. The confidence field prevents uncertain extraction from pretending to be certain observation.

Let Ω denote a fine-grained substrate state space and let ω_{A_t} denote the substrate history over a finite region A_t. The adjective fine-grained describes retained detail and imposes no topology on Ω. A coarse-graining map at temporal resolution Δ_τ:

$$\Gamma_{\Delta_\tau} : \Omega^{A_t} \to \mathcal{E}_{\Delta_\tau}$$

produces an event frame:

$$e_t = \Gamma_{\Delta_\tau}(\omega_{A_t}).$$

This equation states the ontology clearly: the event frame is a lossy, task-oriented compression. The compression is useful only if it preserves distinctions that matter for prediction, intervention, memory, or review.

The temporal resolution Δ_τ controls how precise the "when" field is. A model may choose second-level frames, microsecond-level frames, or another declared scale. Finer resolution can instantiate more candidate frames, but it does not imply that every candidate frame is predictively effective, causally effective, or worth retaining forever. Predictive and causal sparsity are measured separately over a finite declared candidate set; neither is inferred from the size of the substrate.

Mathematically, the event space is treated as a typed product:

$$\mathcal{E} = \mathcal{W} \times \mathcal{A} \times \mathcal{T} \times \mathcal{L} \times \mathcal{M} \times \mathcal{H} \times \mathcal{X} \times \mathcal{C}.$$

This equation is operational, not decorative. It says that before a field can be used in prediction, caching, fuzzing, or abstraction, the field must have a representation and a comparison rule. For example, $\mathcal{T}$ may contain timestamps or intervals; $\mathcal{L}$ may contain coordinates, graph nodes, or symbolic regions; $\mathcal{C}$ may contain confidence scores and source provenance. EventFrame does not require one universal encoding for all domains, but it requires the encoding to be declared.

An event state is the system state before, during, or after an event. In some domains, x_t may include an explicit pre-state and post-state. In others, x_t may be a latent state vector inferred from observations. A transition occurs when one event context gives rise to a later event frame. The basic trajectory is:

$$E_{1:T} = (e_1, e_2, \ldots, e_T).$$

Operationally, a prediction step extracts a context from this trajectory:

$$C_t = e_{t-k+1:t}.$$

The context C_t is the recent event history available to the predictor. It gives the predictor local structure for forecasting the next event and its time. If the context is too short, important predictive or temporal dependencies may be missing. Causal dependence is a stronger claim and requires an explicit causal model or identified intervention evidence. The context length k is a modeling choice that should be evaluated experimentally.

The ontology also supports typed links. Temporal links order events, spatial links relate locations, and predictive-dependency links record forecast-relevant association. Causal links are reserved for relations supported by a declared structural causal model or an identified intervention. These link types must remain distinct in storage and evaluation.

Event histories are therefore not limited to linear chains. Multiple event streams can become representable as a single aggregate event over time. This is event confluence: separate streams merge into a larger stream or macro-event when their separate identities no longer affect the target beyond a declared threshold. The reverse can also occur. A small distinction can branch into multiple downstream event streams when a perturbation is amplified by the dynamics. This is event divergence, or butterfly-effect-style sensitivity. EventFrame must model both patterns because compression that is safe in a confluence region may be unsafe near a divergence point.

For traceability, EventFrame keeps at least one concrete frame for every event-frame group. One frame cannot characterize a heterogeneous group, so abstraction audits use a coverage-aware set containing boundary, uncertain, and sampled examples. Section 7 formalizes that audit set and the limits of conclusions drawn from it.

The event sparsity hypothesis follows from this compression view. Relative to a finite declared candidate set, EventFrame hypothesizes that only a small fraction of distinctions materially worsen held-out proper risk when ablated. This predictive ratio is observationally testable under the fixed ablation protocol. A separate causal ratio requires randomized or otherwise identified interventions. Both must be measured in each domain rather than inferred from Planck constants or entropy bounds.

The main limitation of the ontology is extraction and compression quality. In real data, the "why" and "how" fields may be ambiguous, inferred, or unavailable. More fundamentally, the chosen coarse-graining Γ_{Δ_τ} may discard distinctions that later turn out to matter. EventFrame handles this by allowing missing values, confidence metadata, and revision under slow-path review rather than requiring false precision. A conservative implementation should distinguish observed fields from inferred fields and should propagate uncertainty into prediction and review. The next section defines the mathematical framework built on this compressed ontology.

3. Mathematical Framework

The mathematical framework turns compressed event frames into objects that can be predicted, evaluated, cached, and abstracted. Given a context C_t, the predictor must produce a next-event distribution before the next observation exists. Only after the observation arrives may the runtime compute realized prediction loss and update memory or abstraction.

Let Ω denote a fine-grained substrate state space; fine-grained refers to retained detail and imposes no topology on Ω. For a finite region A_t, let $\omega_{A_t} \in \Omega^{A_t}$ denote the substrate history over that region. At temporal resolution Δ_τ, an event frame is produced by:

$$e_t = \Gamma_{\Delta_\tau}(\omega_{A_t}), \qquad \Gamma_{\Delta_\tau} : \Omega^{A_t} \to \mathcal{E}_{\Delta_\tau}.$$

The coarse-graining Γ_{Δ_τ} is task-relative and lossy. It selects distinctions available to prediction, memory, and review; it does not establish a fundamental discretization of spacetime. For a physical substrate, CODATA Planck scales and physical information bounds motivate a limiting thought experiment about microscopic description [7–9]. They do not imply a Planck-scale sampling lattice or prove EventFrame sparsity. Simulated and software substrates require an independent task- and resource-based compression argument; the physical citations do not support that case.

A trajectory at fixed resolution is:

$$E_{1:T} = (e_1, \ldots, e_T), \qquad e_t \in \mathcal{E}_{\Delta_\tau},$$

and a time quantizer is:

$$Q_{\Delta_\tau} : \mathbb{R} \to \mathcal{T}_{\Delta_\tau}.$$

Second-level or microsecond-level precision is permitted only when the measurement process supports it. Finer resolution creates more candidate frames and can expose boundaries, but also increases noise and cache pressure.

For a context length k, define:

$$C_t = e_{t-k+1:t} \in \mathcal{E}^k.$$

Let $\varnothing \neq \mathfrak{C}_{\mathrm{adm}} \subseteq \mathcal{E}^k$ be the declared admissible context domain on which the chosen version of each conditional forecast law is defined. Population suprema below range over this domain or over the support of a named evaluation law, not over arbitrary zero-probability contexts.

Let $a(x)$ be the time at which observation, label, cache record, or derived object x becomes available to the runtime. Let $\mathscr{F}_t^{\mathrm{pred}}$ be the information available when the prediction at index t is issued, including C_t but excluding Z_{t+1}. Mutable runtime state is the left-limit snapshot S_{t^-}, constructed only from objects with $a(x) \leq t$. Every prediction, priority, cache lookup, abstraction decision, and pre-risk value used at time t must be measurable with respect to $\mathscr{F}_t^{\mathrm{pred}}$. For a no-event outcome, $a(Z_{t+1})$ is no earlier than expiration of horizon H; delayed labels use their actual later availability time.

Let $\nu(e)$ be the event mark or occurrence type. Let $\tau(e) \in \mathbb{R}$ be a declared scalar temporal anchor: it is the timestamp for a point event and, by default, the onset for an interval event. A domain may choose another measurable anchor, such as midpoint, but must freeze that convention before fitting and use it consistently in labeling, caching, and evaluation; the complete interval remains available in the event frame. Over a prediction horizon $H > 0$, the next outcome is:

$$Z_{t+1} = \begin{cases} (\nu(e_{t+1}), \tau(e_{t+1}) - \tau(e_t)), & \text{if an event occurs within } H, \\ \varnothing, & \text{otherwise.} \end{cases}$$

Let $(\mathcal{N}, \mathscr{A}_{\mathcal{N}})$ be the measurable mark space and let $((0, H], \mathscr{B}_H)$ carry the Borel sigma-algebra in the declared time units. Define the marked branch and complete outcome space as the measurable disjoint union

$$(\mathcal{Z}_H^+, \mathscr{A}_H^+) = (\mathcal{N} \times (0, H], \mathscr{A}_{\mathcal{N}} \otimes \mathscr{B}_H), \qquad (\mathcal{Z}_H, \mathscr{A}_H) = (\mathcal{Z}_H^+ \sqcup \{\varnothing\}, \mathscr{A}_H^+ \oplus 2^{\{\varnothing\}}).$$

A probabilistic predictor returns a distribution rather than only a point:

$$\mathsf{Q}_\theta(\cdot \mid C_t) \in \mathcal{P}(\mathcal{Z}_H),$$

where $\mathcal{P}(\mathcal{Z}_H)$ denotes probability measures on $(\mathcal{Z}_H, \mathscr{A}_H)$, equipped with the evaluation sigma-algebra generated by $Q \mapsto Q(A)$ for $A \in \mathscr{A}_H$. This is a finite-horizon marked-event representation with a right-censoring atom; standard marked point-process constructions provide a broader continuous-time setting [11]. Let $\mathcal{E}_\varnothing = \mathcal{E} \sqcup \{\varnothing\}$ be the tagged extension of the structured event space. A fixed measurable decision rule

$$d_H : \mathcal{P}(\mathcal{Z}_H) \to \mathcal{Z}_H$$

returns the no-event decision or a marked time, with a declared loss and deterministic tie-break. Let $\hat{e}^H_\theta(C_t) \in \mathcal{E}_\varnothing$ be a structured summary coherent with that decision: it equals $\varnothing$ exactly when $d_H(\mathsf{Q}_\theta(\cdot \mid C_t)) = \varnothing$, and otherwise its mark and time agree with the marked decision. The typed predictor output is the bundle:

$$\mathcal{O}_\theta(C_t) = \big(\mathsf{Q}_\theta(\cdot \mid C_t), \hat{e}^H_\theta(C_t)\big) \in \mathcal{P}(\mathcal{Z}_H) \times \mathcal{E}_\varnothing.$$

The primary probabilistic-fidelity objective is a declared strictly proper scoring rule applied to the probability-law component. It is distinct from the composite system-design objective below:

$$\mathcal{L}_{\text{pred}}(\theta; t) = S_{\text{prop}}(\mathsf{Q}_\theta(\cdot \mid C_t), Z_{t+1})\,.$$

Fix a dominating reference measure μ_H on the marked-time branch, including declared units for time, and let $q_\theta = d\mathsf{Q}^{\text{event}}_\theta / d\mu_H$ be the event subdensity. Its integral equals $1 - \mathsf{Q}_\theta(\{\varnothing\} \mid C_t)$. Relative to this fixed reference measure, the logarithmic score is one implementation:

$$\mathcal{L}_{\log}(\theta; t) = \begin{cases} -\log q_\theta(\nu_{t+1}, \Delta t_{t+1} \mid C_t), & Z_{t+1} \neq \varnothing, \\ -\log \mathsf{Q}_\theta(\{\varnothing\} \mid C_t), & Z_{t+1} = \varnothing. \end{cases}$$

If $d\mu'_H = h\, d\mu_H$, then the marked-branch logarithmic loss changes by the forecast-independent but generally outcome-dependent term $\log h(Z)$. A pure unit rescaling makes this term constant. Such an integrable outcome-only term preserves propriety and pairwise forecast differences on the same observation, but the reference measure and units must remain fixed when absolute scores or results across datasets are reported. This score covers event identity, timing, uncertainty, and right-censoring; calibration remains an empirical property to test. Proper scoring rules prevent a predictor from improving its expected score by reporting a distribution other than the one it believes [6].

For human-readable diagnostics, let $\hat{Z}_{t+1} = d_H(\mathsf{Q}_\theta(\cdot \mid C_t))$. A bounded event-aware timing diagnostic is:

$$\mathcal{L}^H_{\text{event}}(\hat{Z}, Z) = \begin{cases} 0, & \hat{Z} = Z = \varnothing, \\ 1, & \text{exactly one is } \varnothing \text{ or their marks differ}, \\ \min\left(1, \dfrac{|\widehat{\Delta t} - \Delta t|}{H}\right), & \text{their non-null marks agree.} \end{cases}$$

Unlike the original timing-only diagnostic, this expression cannot assign zero loss to the wrong event type merely because its timestamp is correct. It remains a diagnostic; model fitting and forecast comparison should use $\mathcal{L}_{\text{pred}}$.

For any other field, use a distinct projection $\psi_i : \mathcal{E} \to \mathcal{X}_i$ and declared distance. Bind the coherent structured prediction by $\hat{e}^H_{\theta,t+1} := \hat{e}^H_\theta(C_t)$. The ordinary field loss is defined only when both the decision and observation are marked events:

$$\mathcal{L}_i(\hat{e}^H_{\theta,t+1}, Z_{t+1}) = d_i(\psi_i(\hat{e}^H_{\theta,t+1}), \psi_i(e_{t+1})), \qquad \hat{e}^H_{\theta,t+1} \in \mathcal{E}, \quad Z_{t+1} \neq \varnothing.$$

When either argument is $\varnothing$, this field loss is not evaluated unless a separate missing-aware loss on $\mathcal{E}_\varnothing$ has been declared. The proper score and event diagnostic still evaluate the no-event decision.

EventFrame uses separate pre-observation and post-observation quantities. For a candidate output bundle $\widetilde{\mathcal{O}} = (\widetilde{\mathsf{Q}}, \tilde{e}^H)$, a pre-observation admissibility risk may use only information available at prediction time:

$$\mathcal{R}_{\text{pre}}(\widetilde{\mathcal{O}} \mid C_t) = \lambda_a^{\text{pre}} D_{\text{abs}}^{\text{pre}}(\widetilde{\mathcal{O}}, C_t) + \lambda_c^{\text{pre}} D_{\text{edge}}^{\text{pre}}(\widetilde{\mathcal{O}}, C_t) + \lambda_u^{\text{pre}} U^{\text{pre}}(\widetilde{\mathcal{O}} \mid C_t).$$

The three components lie in $[0, 1]$, the weights are non-negative, and $\lambda_a^{\text{pre}} + \lambda_c^{\text{pre}} + \lambda_u^{\text{pre}} = 1$.

The proper loss need not be bounded; the logarithmic score above equals $+\infty$ when the realized outcome receives zero density or mass. To combine an extended-real score with bounded system diagnostics, choose and preregister a measurable non-decreasing map $g_{\text{pred}} : \overline{\mathbb{R}} \to [0, 1]$ on the score's entire attainable extended-real range, including explicit endpoint values. For a non-negative log loss, one admissible example is

$$g_{\text{pred}}(\ell) = 1 - \exp(-\ell/\kappa), \quad \ell \in [0, \infty), \qquad g_{\text{pred}}(+\infty) = 1, \quad \kappa > 0.$$

For a score with negative attainable values, the evaluation contract instead supplies a total monotone transform on that stated range, including $-\infty$ if attainable. Define:

$$\overline{\mathcal{L}}_{\text{pred}}(\widetilde{\mathsf{Q}}, Z) = g_{\text{pred}}\Big(S_{\text{prop}}(\widetilde{\mathsf{Q}}, Z)\Big).$$

Constant or order-reversing transforms are inadmissible. Unless g_{pred} is a positive affine transformation on the score's range, $\overline{\mathcal{L}}_{\text{pred}}$ is not asserted to remain proper. Model fitting and forecast comparison continue to report the untransformed proper score. After Z_{t+1} is observed, the bounded realized event action is:

$$\begin{aligned}\mathcal{A}_{\text{post}}(\widetilde{\mathcal{O}}, Z_{t+1}) = {} & \lambda_p^{\text{post}} \overline{\mathcal{L}}_{\text{pred}}(\widetilde{\mathsf{Q}}, Z_{t+1}) + \lambda_a^{\text{post}} D_{\text{abs}}^{\text{post}}(\widetilde{\mathcal{O}}, Z_{t+1}) \\ & + \lambda_c^{\text{post}} D_{\text{edge}}^{\text{post}}(\widetilde{\mathcal{O}}, Z_{t+1}) + \lambda_u^{\text{post}} U^{\text{post}}(\widetilde{\mathcal{O}}, Z_{t+1}).\end{aligned}$$

Every post-observation component lies in $[0, 1]$, the four post weights are non-negative and sum to one, and therefore $\mathcal{A}_{\text{post}} \in [0, 1]$.

The evaluation contract supplies the complete measurable definitions of $D_{\text{abs}}^{\text{pre}}$, $D_{\text{edge}}^{\text{pre}}$, U^{pre}, their post-observation counterparts, and every target, normalization, missing-value rule, threshold, and weight they use. Alternatively it may supply finite admissible classes plus a design-sample fitting and deterministic tie-breaking procedure. Packet-target construction, packet component loss, priority-model class and fitting rule, and regime-shift detection window, repetition rule, threshold, and resulting action are frozen under the same requirement. None may be selected or retuned from confirmation outcomes or separately for the candidate it scores. A learned component is fitted using only the designated design history, charged to representation or fitting cost when appropriate, and frozen before confirmation.

The fast path may gate a correction using $\mathcal{R}_{\text{pre}}$; it may never use $\mathcal{A}_{\text{post}}$ before the observation exists.

The governing principle can now be stated without overloading Ω. It is evaluated at a fixed resolution Γ_{Δ_τ}; comparisons across resolutions are a separate outer experiment on common raw histories. Group the event-residual implementation contract as:

$$\Xi_R = (q_E, d_E, \Pi_E, \delta_E, \delta_Q, \mathcal{M}_R, \rho_H^Q, \mathfrak{K}_H^Q, d_H, \mathrm{lift}_H, \alpha, \kappa, \epsilon_R, \text{cache gates}),$$

and the candidate abstraction structure as $\Xi_A^{(v)}$, containing a versioned compatibility graph, its assigned comparison spaces and maps, and the declared edge divergences and weights. The version v changes only when a validated slow-path revision is published.

The selective Bayesian contract Ξ_B contains the bounded vector, sheaf-inspired, and as-of graph frontier rules; activation and criticality maps; posterior family and likelihood; selection model; Anti-Pigeon sharing certificate; source-dependence treatment; changepoint approximation; independent audit schedule; omitted-influence test; resource caps; and atomic publication rule. Its as-of posterior cache is $\mathcal{C}_{B,t^-}$. Outgoing graph relationships may nominate candidates but cannot supply evidence about outcomes that have not yet become available. Separately freeze an evaluation contract:

$$\begin{aligned}\Lambda_{\text{eval}} = (&P_{\text{obj}}, P_{\text{conf}}, \mathcal{S}_{\text{obj}}, \mathcal{S}_{\text{conf}}, P_\star, \mathfrak{C}_{\text{adm}}, d_C, \text{targets, divergences, thresholds,}\\ &\text{complete diagnostics or finite admissible classes, fitting and tie-break rules,}\\ &g_{\text{pred}}, \text{score weights}, p^{\text{pri}}, w_{\text{pri}}, \lambda_{\text{rep}}, \mathcal{C}_{\text{rep}},\\ &\text{packet target and loss, regime-shift rule, confidence and map-validity procedures,}\\ &\text{snap candidate, obligation, and publication rules,}\\ &\text{Bayesian activation, selection, audit, changepoint, cap, and publication rules).}\end{aligned}$$

Let $\mathfrak{h}_t \in \mathfrak{H}_t$ denote the complete observable history available at prediction origin t, including timestamps and object-availability metadata but excluding Z_{t+1}. The context extractor is $c_k(\mathfrak{h}_t) = C_t$. For each complete design Θ, a measurable deterministic as-of replay operator reconstructs its candidate-specific mutable state:

$$S_{\Theta,t^-} = \mathrm{Replay}_\Theta(\mathfrak{h}_t).$$

Here P_{obj} and P_{conf} are fixed design- and confirmation-generating laws on prediction instances $(\mathfrak{h}_t, Z_{t+1})$. Their realized, chronologically separated samples or trajectory blocks are $\mathcal{S}_{\text{obj}} \sim P_{\text{obj}}$ and $\mathcal{S}_{\text{conf}} \sim P_{\text{conf}}$; independence is not assumed unless the sampling design supplies it. Replaying each candidate on the same raw history permits its caches, posteriors, epochs, changepoint states, confidence, and prior updates to differ without treating state as an unintegrated free variable. The external target conditional law is $P_\star(Y \mid C)$. This contract is fixed independently of the candidates being compared; a candidate cannot shrink the history or context domain, relax its thresholds, choose its own weights, redefine the target, or validate its own comparison maps. Require $\lambda_{\text{rep}} \geq 0$ and $\mathcal{C}_{\text{rep}} \geq 0$. At the fixed resolution, let:

$$\Theta_\Gamma = (\mathsf{Q}_\theta, B, \pi, \mathcal{C}_A, \mathcal{C}_R, \mathcal{C}_E, \mathcal{C}_B, \Xi_R, \Xi_B, \Xi_A^{(v)})$$

denote the complete event-prediction design evaluated under Λ_{eval}. Let $\mathcal{O}_{\Theta_\Gamma}(C; S_{\Theta_\Gamma,t^-})$ denote its final typed output bundle from the candidate state reconstructed immediately before prediction time. Let $\mathfrak{K}_\pi$ be the buckets induced by π, and let $\mathfrak{K}_\pi^+ = \{K \in \mathfrak{K}_\pi : \mathfrak{C}_K \neq \varnothing\}$ be the active buckets with admissible contexts. For an active bucket K, define its external future-diameter $D_K^\star(\pi)$ as in Section 7 under the fixed target law, divergence, and context domain. Runtime-packet contracts are evaluated by their separate packet loss and are added to Θ_Γ only in an implementation that jointly optimizes packet selection.

Compression must be operational, not merely decorative. Define retained information by

$$h_\pi(C) = \big(\pi^{(k)}(C), s_\pi(C)\big),$$

where s_π is declared side information. There must exist measurable maps $\widetilde{\mathsf{Q}}_\theta, \widetilde{B}, \widetilde{\alpha}, \widetilde{\kappa}$ such that $\mathsf{Q}_\theta = \widetilde{\mathsf{Q}}_\theta \circ h_\pi$, $B = \widetilde{B} \circ h_\pi$, $\alpha = \widetilde{\alpha} \circ h_\pi$, and $\kappa = \widetilde{\kappa} \circ h_\pi$. The storage and acquisition cost of s_π is charged to $\mathcal{C}_{\mathrm{rep}}$. Without this factorization, π may remain an interpretive annotation, but the system must not claim operational compression through π.

Let $p^{\mathrm{pri}}(C; S_{\Theta,t^-}) \in [0,1]$ be priority assigned from information available at prediction time and let $w_{\mathrm{pri}}(p) > 0$ be a declared importance function with finite, positive mean. The priority model, its preprocessing, and the weight function are fitted only on data available before the evaluated block and are frozen independently of the candidates. The unweighted objective is recovered by setting $w_{\mathrm{pri}} \equiv 1$. For $D \in \{P_{\mathrm{obj}}, P_{\mathrm{conf}}\}$, define trajectory-instance risk:

$$\mathcal{R}^D_{\mathrm{pri}}(\Theta_\Gamma) = \frac{\mathbb{E}_{(\mathfrak{h}_t, Z)\sim D}\left[w_{\mathrm{pri}}(p^{\mathrm{pri}}(c_k(\mathfrak{h}_t); S_{\Theta_\Gamma,t^-}))\mathcal{A}_{\mathrm{post}}(\mathcal{O}_{\Theta_\Gamma}(c_k(\mathfrak{h}_t); S_{\Theta_\Gamma,t^-}), Z)\right]}{\mathbb{E}_{\mathfrak{h}_t\sim D}\left[w_{\mathrm{pri}}(p^{\mathrm{pri}}(c_k(\mathfrak{h}_t); S_{\Theta_\Gamma,t^-}))\right]}.$$

Let S_{prop} be a predeclared strictly proper scoring rule on the predictive-law component, and define the unweighted proper risk

$$\mathcal{R}^D_{\mathrm{prop}}(\Theta_\Gamma) = \mathbb{E}_{(\mathfrak{h}_t, Z)\sim D}\left[S_{\mathrm{prop}}(\mathsf{Q}_{\Theta_\Gamma}(\cdot \mid c_k(\mathfrak{h}_t); S_{\Theta_\Gamma,t^-}), Z)\right].$$

All displayed population expectations must be finite. A candidate with infinite empirical proper loss is assigned an infinite proper-risk guard statistic and is not certifiable; numerical probability flooring is permitted only if it is part of the frozen forecast family and its effect on propriety is stated. The proper risk prevents improvements in a bounded composite score from being purchased by a worse probabilistic forecast.

The population target law defines an oracle benchmark, not an implementable selector. Define the oracle feasible family:

$$\mathfrak{F}_{AP}^{\Gamma,\star} = \left\{\Theta_\Gamma : \begin{array}{l} D_K^\star(\pi) \le \epsilon_{AP} \text{ for every } K \in \mathfrak{K}_\pi^+, \\ \text{the operational factorization through } h_\pi \text{ holds}, \\ \mathcal{R}^{P_{\mathrm{obj}}}_{\mathrm{prop}}(\Theta_\Gamma) \le \mathcal{R}^{P_{\mathrm{obj}}}_{\mathrm{prop}}(\Theta_{\Gamma,0}) + \epsilon_{\mathrm{prop}} \end{array}\right\}.$$

Here $\Theta_{\Gamma,0}$ is a frozen reference predictor and $\epsilon_{\mathrm{prop}} \ge 0$ is declared in advance. The oracle governing value is:

$$\boxed{\mathcal{J}_\Gamma^{\mathrm{oracle}} = \inf_{\Theta_\Gamma \in \mathfrak{F}_{AP}^{\Gamma,\star}} \left[\mathcal{R}^{P_{\mathrm{obj}}}_{\mathrm{pri}}(\Theta_\Gamma) + \lambda_{\mathrm{rep}}\mathcal{C}_{\mathrm{rep}}(\Theta_\Gamma)\right]}.$$

This value states the desired population property. When $P_\star$ is unknown it is not directly computable, and no finite-sample algorithm may claim membership in $\mathfrak{F}_{AP}^{\Gamma,\star}$ merely from a small point estimate. If the oracle feasible set is non-empty and its infimum is attained, an oracle optimizer satisfies:

$$\Theta_\Gamma^{\text{oracle}} \in \arg\min_{\Theta_\Gamma \in \mathfrak{F}_{AP}^{\Gamma,\star}} \Big[\mathcal{R}_{\text{pri}}^{P_{\text{obj}}}(\Theta_\Gamma) + \lambda_{\text{rep}} \mathcal{C}_{\text{rep}}(\Theta_\Gamma) \Big] .$$

Operational selection instead begins with a finite, predeclared candidate family $\mathfrak{G}_\Gamma(\mathcal{S}_{\text{obj}})$, constructed using design data only. A candidate is certifiable only when every active bucket has either exhaustive audit coverage or the verified continuity certificate from Section 7. Let $\widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{obj}}}$ and $\widehat{\mathcal{R}}_{\text{pri}}^{\mathcal{S}_{\text{obj}}}$ be the corresponding grouped, as-of empirical risks, and define:

$$\widehat{\mathfrak{F}}_{AP}^{\Gamma} = \left\{ \Theta_\Gamma \in \mathfrak{G}_\Gamma(\mathcal{S}_{\text{obj}}) : \begin{array}{l} D_K^{\text{cert},\star}(\pi) \le \epsilon_{AP} \text{ for every } K \in \mathfrak{K}_\pi^+, \\ \text{the operational factorization through } h_\pi \text{ holds,} \\ \text{UCB}\Big[\widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{obj}}}(\Theta_\Gamma) - \widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{obj}}}(\Theta_{\Gamma,0}) \Big] \le \epsilon_{\text{prop}} \end{array} \right\} .$$

If $\widehat{\mathfrak{F}}_{AP}^{\Gamma} \neq \varnothing$, the implementable design rule is the finite minimum

$$\widehat{\Theta}_\Gamma \in \arg\min_{\Theta_\Gamma \in \widehat{\mathfrak{F}}_{AP}^{\Gamma}} \Big[\widehat{\mathcal{R}}_{\text{pri}}^{\mathcal{S}_{\text{obj}}}(\Theta_\Gamma) + \lambda_{\text{rep}} \mathcal{C}_{\text{rep}}(\Theta_\Gamma) \Big] ,$$

with a declared deterministic tie-break. If the certified family is empty, the procedure returns no admissible design or a separately declared conservative fallback; it does not relax the thresholds. The confidence guarantees apply to the stated finite-sample constraints, not to attainment of the oracle infimum.

Selection and tuning use only $\mathcal{S}_{\text{obj}}$. After the candidate, preprocessing, thresholds, priority rule, and analysis are frozen, final claims are evaluated once on untouched $\mathcal{S}_{\text{conf}}$. Both samples use rolling-origin or forward-chaining construction under their named generating laws, grouped by independent trajectory or entity where applicable, with an embargo long enough to cover context overlap, forecast horizon, and label delay. Weighted results are accompanied by unweighted and priority-stratified results. Oracle feasibility does not guarantee empirical certifiability, and empirical certifiability does not prove unrestricted population feasibility beyond the certificate's assumptions and coverage. Cross-resolution comparisons use the same raw histories and fixed target law; a candidate resolution may not redefine the outcome it is judged against.

An event history may be represented by a time-unrolled directed graph:

$$G_t = (V_t, R_t),$$

where $V_t \subset \mathcal{E}$ and edges in R_t are typed as temporal, predictive-dependency, or causal. The graph is acyclic only after time-unrolling; feedback in the physical system is represented through edges across successive times. Predictive-dependency edges must not be interpreted as causal edges without a structural causal model.

For causal language, EventFrame requires an explicit structural causal model $\mathfrak{M} = (U, V, F, P_U)$. An intervention such as $do(V_j = v')$ replaces the structural equation for V_j; only then is

$$\Delta_Y^{\text{causal}}(v'; P_{\text{ref}}) = D_Y^{\text{law}}\big(P_{\mathfrak{M}}(Y \mid do(V_j = v')), P_{\text{ref}}(Y)\big)$$

a causal effect magnitude relative to a declared reference law P_{ref}, such as the natural-course law $P_{\mathfrak{M}}(Y)$ or another intervention law [5]. It is not a signed effect, and its interpretation depends on the chosen distance and reference. Without $\mathfrak{M}$, changing an input frame or graph is a model perturbation and measures predictor sensitivity, not causation.

The event sparsity hypothesis is stated relative to a finite, non-empty declared candidate set $\mathcal{D}_t$, not by comparing cardinalities with a continuous substrate. Predictive and causal relevance are different estimands. For each distinction $d \in \mathcal{D}_t$, let Θ_Γ^{-d} be a predeclared ablation or coarsening fitted on design data under the same protocol. Define the paired proper-risk effect under the confirmation-generating law and, after all full and ablated designs are frozen, its empirical estimate:

$$\Delta_{\text{pred}}(d) = \mathcal{R}_{\text{prop}}^{P_{\text{conf}}}(\Theta_\Gamma^{-d}) - \mathcal{R}_{\text{prop}}^{P_{\text{conf}}}(\Theta_\Gamma), \qquad \widehat{\Delta}_{\text{pred}}(d) = \widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{conf}}}(\Theta_\Gamma^{-d}) - \widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{conf}}}(\Theta_\Gamma).$$

With a predeclared threshold $\eta_{\text{pred}} \geq 0$, define the observationally evaluable predictive ratio using a paired simultaneous confidence procedure over the entire declared distinction family:

$$s_{\text{eff}}^{\text{pred}} = \frac{|\{d \in \mathcal{D}_t : \text{LCB}_{\text{sim}}[\Delta_{\text{pred}}(d)] > \eta_{\text{pred}}\}|}{|\mathcal{D}_t|}.$$

The confidence bound is constructed from $\widehat{\Delta}_{\text{pred}}(d)$. Confirmation outcomes may classify the frozen distinctions but may not refit, regenerate, or select the candidate family. This is a predictive association under the fixed ablation and evaluation distribution, not a causal effect. If $\mathcal{I}_{\text{eff}}^{\text{causal}}(Y, \eta_Y) \subseteq \mathcal{D}_t$ contains distinctions whose randomized or otherwise identified intervention-effect magnitude exceeds η_Y, define separately:

$$s_{\text{eff}}^{\text{causal}} = \frac{|\mathcal{I}_{\text{eff}}^{\text{causal}}(Y, \eta_Y)|}{|\mathcal{D}_t|}.$$

EventFrame hypothesizes $s_{\text{eff}}^{\text{pred}} \ll 1$ in domains where predictive compression is useful. It may hypothesize $s_{\text{eff}}^{\text{causal}} \ll 1$ only in a domain where the required interventions are identified. Both are falsifiable domain-level hypotheses, not physical theorems.

Confluence and divergence concern target-relative predictive behavior. A merge $\mu_\delta(S_1, \ldots, S_m)$ is accepted only when its held-out predictive degradation and bucket future-diameter remain below declared thresholds. A perturbation operator $\mathcal{B}_\epsilon$ may generate candidate downstream graphs, but a distribution over those candidates must be specified before writing probabilities conditioned on its output.

Every non-empty event bucket K retains at least one concrete frame $\bar{e}_K \in K$ for traceability. Future-divergence detection audits contexts, because the same frame may occur after different histories. With $\text{anc}(C)$ denoting the terminal frame of context C, let $\mathfrak{C}_K = \{C \in \mathfrak{C}_{\text{adm}} : \text{anc}(C) \in K\}$. For an active bucket, maintain a non-empty $\mathcal{R}_C(K) \subseteq \mathfrak{C}_K$ satisfying a declared context-coverage rule, for example:

$$\sup_{C \in \mathfrak{C}_K} \min_{R \in \mathcal{R}_C(K)} d_C(C, R) \le \delta_K.$$

The audit set may combine contexts for a medoid, boundary examples, high-uncertainty examples, and a reservoir sample. Tests over $\mathcal{R}_C(K)$ are statistical estimates, not proofs about unobserved contexts. A certified future-diameter bound additionally requires exhaustive coverage or the verified continuity condition in Section 7. Confidence, coverage, and false-negative risk must be reported.

Confidence and provenance metadata c_t determine whether fields may be used for training, lookup, sensitivity testing, or causal analysis. Observed fields, inferred fields, and synthetic perturbations remain distinct throughout the lifecycle.

4. Residual Prediction

Residual prediction separates a first-pass event estimate from a correction. The baseline captures ordinary transition structure; the residual records a recurring statistical prediction error. A residual is not a causal hypothesis unless separate intervention evidence identifies it as causal.

Let the baseline probability law and its conditional structured event template be:

$$\mathsf{Q}_B : \mathcal{E}^k \to \mathcal{P}(\mathcal{Z}_H), \qquad B : \mathcal{E}^k \to \mathcal{E}, \qquad b_t = B(C_t).$$

To make structured correction type-correct, choose a finite-dimensional Hilbert space $\mathscr{H}$. Let $\mathbb{H}_d^E$ and $\mathbb{H}_d^Q$ be separately tagged copies of the real vector space of self-adjoint operators on $\mathscr{H}$, each equipped with the Frobenius norm $\|\cdot\|_F$. The superscripts distinguish point-template semantics from forecast-law semantics even when an implementation uses the same matrix representation. Define:

$$q_E : \mathcal{E} \to \mathbb{H}_d^E, \qquad d_E : \mathcal{Q}_{E,\mathrm{adm}} \to \mathcal{E},$$

where $\mathcal{Q}_{E,\mathrm{adm}} \subseteq \mathbb{H}_d^E$ is a non-empty closed admissible set and d_E is a decoder, not an inverse of the lossy encoder. For a radius $\delta_E > 0$, define point-residual norm clipping by:

$$\mathrm{clip}_{\delta_E}(r) = \begin{cases} 0, & r = 0, \\ r \min\left(1, \dfrac{\delta_E}{\|r\|_F}\right), & r \neq 0. \end{cases}$$

Let the admissibility projection be a deterministic selection:

$$\Pi_E(v) \in \arg\min_{u \in \mathcal{Q}_{E,\mathrm{adm}}} \|u - v\|_F.$$

A minimizer exists when the admissible set is closed in finite dimensions. It is unique when that set is convex; otherwise the implementation must declare a tie-breaking rule. Event residual composition is:

$$b \oplus_E r = \begin{cases} b, & r = 0, \\ d_E\big(\Pi_E\big(q_E(b) + \mathrm{clip}_{\delta_E}(r)\big)\big), & r \neq 0, \end{cases} \qquad r \in \mathbb{H}_d^E.$$

Thus zero is an exact identity even when the encoder is lossy or the baseline encoding is outside the admissible set.

This construction takes limited inspiration from the use of self-adjoint operator representations in Causal Fermion Systems [1,2]. The clipping radius, admissible set, projection, decoder, and residual objective are EventFrame definitions; they are not CFS terminology or consequences of the CFS causal action principle. The construction does not inherit CFS field equations and makes no claim of physical equivalence.

Residuals are estimated after observation and are indexed by the forecast horizon that generated their label. For the forecast issued at t, a simple point-representation residual exists only when the concrete next event lies inside that same horizon:

$$r_{t,H}^{E,\mathrm{obs}} = q_E(e_{t+1}) - q_E(b_t), \qquad 0 < \tau(e_{t+1}) - \tau(e_t) \leq H.$$

If $Z_{t+1} = \varnothing$, then $r_{t,H}^{E,\mathrm{obs}}$ is undefined for that forecast origin. A concrete event observed after H may label a later forecast origin, but it must not retroactively become the point residual of the expired H-horizon forecast. To learn a law correction from either branch, declare a measurable horizon-specific distributional residual estimator with a separately tagged codomain

$$\rho_H^Q : \mathcal{P}(\mathcal{Z}_H) \times \mathcal{Z}_H \to \mathbb{H}_d^Q, \qquad r_{t,H}^{Q,\mathrm{obs}} = \rho_H^Q(\mathsf{Q}_B(\cdot \mid C_t), Z_{t+1}).$$

Its objective may be a proper-score gradient, a constrained law update, or another predeclared rule, but it must be defined at $Z_{t+1} = \varnothing$, fitted without future leakage, and evaluated on later outcomes. Define the residual mode set $\mathcal{M}_R = \{\varnothing, E, Q, EQ\}$ and a typed residual record

$$\mathbf{r} = (r^E, r^Q, m) \in \mathbb{H}_d^E \times \mathbb{H}_d^Q \times \mathcal{M}_R.$$

Write $\mathcal{V}_R = \mathbb{H}_d^E \times \mathbb{H}_d^Q \times \mathcal{M}_R$ and $\mathbf{0}_R = (0_E, 0_Q, \varnothing)$.

The mode says which components are semantically present; an absent component is stored as zero, but zero remains a valid present correction when its mode includes that component. A point-only record $m = E$ may support point diagnostics but cannot change the forecast law. A law-only record $m = Q$ may change the law and proper score while leaving non-mark, non-time template fields at baseline. A joint record $m = EQ$ contains separately estimated components and must pass joint forward validation of the resulting bundle; it does not assert $r^E = r^Q$ or infer one component's semantics from the other. A no-event observation must not be silently encoded as a concrete point residual. Every cache entry records its mode, estimator identities, horizon, and censoring convention; reuse across horizons requires a separately validated transport rule. In all cases, a stored record is a reusable correction candidate whose utility must be re-evaluated on later observations.

The general residual cache available immediately before prediction is:

$$\mathcal{C}_{R,t^-} = \{(\kappa_i, \mathbf{r}_i, c_i, n_i, t_i, v_i, \mu_i, H_i, s_i)\}_{i=1}^{N_t}, \qquad \kappa : \mathcal{E}^k \to \mathcal{K}_R, \qquad \mathbf{r}_i \in \mathcal{V}_R,$$

where c_i is residual confidence, n_i is effective support, t_i is the last certified update time, v_i is its abstraction epoch, μ_i is its compatibility safety margin, H_i is its forecast horizon, and s_i is provenance including component modes, estimator identities, censoring convention, and eligible training interval. Only entries whose availability time is at most t may occur in $\mathcal{C}_{R,t^-}$. For $N_t > 0$, let:

$$j_t = \min\left(\arg\min_{1\le i\le N_t} d_{\mathcal{K}_R}(\kappa(C_t), \kappa_i)\right),$$

where the outer minimum is the declared deterministic tie-break. Define general-cache acceptance without dereferencing an empty cache:

$$J_t^R = \begin{cases} \mathbf{1}\left[\begin{array}{l} d_{\mathcal{K}_R}(\kappa(C_t), \kappa_{j_t}) \le \epsilon_R, \quad c_{j_t} \ge \gamma_R, \quad n_{j_t} \ge n^R_{\min}, \\ \mathrm{age}_t(t_{j_t}) \le A^R_{\max}, \quad v_{j_t} = v_t, \quad H_{j_t} = H, \quad \mu_{j_t} \ge 0, \quad s_{j_t} \text{ is valid} \end{array}\right], & N_t > 0, \\ 0, & N_t = 0. \end{cases}$$

The retrieved residual is:

$$\mathbf{r}_t^* = \begin{cases} \mathbf{r}_{j_t}, & J_t^R = 1, \\ \mathbf{0}_R, & \text{otherwise.} \end{cases}$$

A valid zero residual is distinguishable from a miss because J_t^R, not its value, records acceptance. Realized loss and cache updates wait for an available Z_{t+1}; the final selector and pre-observation gate are defined only after the candidate bundle below exists.

For lower-latency exact-key reuse, let:

$$\alpha : \mathcal{E}^k \to \mathcal{K}_A,$$

and define the partial map:

$$\mathcal{C}_{A,t^-} : \mathcal{K}_A \rightharpoonup \mathcal{V}_R \times [0,1] \times \mathbb{N}_0 \times \mathcal{T} \times \mathbb{N}_0 \times \mathbb{R} \times \mathbb{R}_{>0} \times \mathcal{S}_{\mathrm{prov}}.$$

For $k_t = \alpha(C_t)$, bind the cache entry only when it exists:

$$k_t \in \mathrm{dom}(\mathcal{C}_{A,t^-}) \implies \mathcal{C}_{A,t^-}(k_t) = (\mathbf{r}_{k_t}, c_{k_t}, n_{k_t}, t_{k_t}, v_{k_t}, \mu_{k_t}, H_{k_t}, s_{k_t}),$$

where n_{k_t} is effective support after accounting for clustered or overlapping trials, v_{k_t} is the cache entry's local abstraction epoch, v_t is the active as-of epoch for the same dependency region, μ_{k_t} is the compatibility safety margin materialized by the slow path, H_{k_t} is the horizon under which the residual was estimated, and s_{k_t} records component modes, estimator identities, censoring convention, and eligible training interval. If $E(k_t)$ is the declared set of compatibility edges on which the entry depends, for example:

$$\mu_{k_t} = \begin{cases} \epsilon^{\mathrm{comp}}_{\mathrm{merge}}, & E(k_t) = \varnothing, \\ \epsilon^{\mathrm{comp}}_{\mathrm{merge}} - \max_{e \in E(k_t)} \mathrm{UCB}_{\mathrm{sim}}[\delta_e], & E(k_t) \neq \varnothing. \end{cases}$$

The simultaneous confidence procedure covers every edge inspected for that cache certificate.

Define the exact-cache acceptance indicator without dereferencing a missing entry:

$$J^A_t = \begin{cases} \mathbf{1}\begin{bmatrix} c_{k_t} \geq \gamma_A, & n_{k_t} \geq n_{\min}, & \mathrm{age}_t(t_{k_t}) \leq A_{\max}, \\ v_{k_t} = v_t, & H_{k_t} = H, & \mu_{k_t} \geq 0, & s_{k_t} \text{ is valid} \end{bmatrix}, & k_t \in \mathrm{dom}(\mathcal{C}_{A,t^-}), \\ 0, & k_t \notin \mathrm{dom}(\mathcal{C}_{A,t^-}). \end{cases}$$

Then:

$$\mathbf{r}^A_t = \begin{cases} \mathbf{r}_{k_t}, & J^A_t = 1, \\ \mathbf{0}_R, & \text{otherwise.} \end{cases}$$

A valid zero residual is now distinguishable from a miss because J^A_t, not the residual value, records acceptance. The exact-to-general selection is:

$$\mathbf{r}^{\mathrm{use}}_t = \begin{cases} \mathbf{r}^A_t, & J^A_t = 1, \\ \mathbf{r}^*_t, & J^A_t = 0 \text{ and } J^R_t = 1, \\ \mathbf{0}_R, & \text{otherwise.} \end{cases}$$

To connect a law residual to the probability law evaluated by the proper score, choose $\delta_Q > 0$, define clip_{δ_Q} on $\mathbb{H}^Q_d$ by the same norm-clipping rule as above, and let $\mathrm{Ker}(\mathcal{Z}_H)$ denote Markov kernels on the measurable space $(\mathcal{Z}_H, \mathscr{A}_H)$. Declare:

$$\mathfrak{K}^Q_H : \mathbb{H}^Q_d \to \mathrm{Ker}(\mathcal{Z}_H), \qquad \mathfrak{K}^Q_H(0_Q)(z, A) = \mathbf{1}_A(z).$$

For every $A \in \mathscr{A}_H$, the evaluation map $(r^Q, z) \mapsto \mathfrak{K}^Q_H(r^Q)(z, A)$ must be jointly measurable on $\mathbb{H}^Q_d \times \mathcal{Z}_H$; for fixed r^Q, it must be a Markov kernel. These conditions supply the measurable structure actually used below without requiring an unspecified sigma-algebra on a function space. The declaration covers every $z \in \mathcal{Z}_H$, including $\varnothing$, for every effective residual. The implementation must explicitly specify both $\mathfrak{K}^Q_H(\bar{r}^Q)(\varnothing, \{\varnothing\})$ and $\mathfrak{K}^Q_H(\bar{r}^Q)(z, \{\varnothing\})$ for $z \in \mathcal{Z}^+_H$; preservation of the no-event atom is not a default assumption.

For $\mathbf{r} = (r^E, r^Q, m)$, define the effective components

$$\bar{r}^E = \begin{cases} \mathrm{clip}_{\delta_E}(r^E), & m \in \{E, EQ\}, \\ 0_E, & m \in \{\varnothing, Q\}, \end{cases} \qquad \bar{r}^Q = \begin{cases} \mathrm{clip}_{\delta_Q}(r^Q), & m \in \{Q, EQ\}, \\ 0_Q, & m \in \{\varnothing, E\}. \end{cases}$$

For every $A \in \mathscr{A}_H$, define:

$$\mathsf{Q}_t^{(\mathbf{r})}(A \mid C_t) = \int_{\mathcal{Z}_H} \mathfrak{K}_H^Q(\bar{r}^Q)(z, A)\, \mathsf{Q}_B(dz \mid C_t).$$

Because $\mathfrak{K}_H^Q(r^Q)$ is a Markov kernel, $\mathsf{Q}_t^{(\mathbf{r})}$ is a probability law. Its no-event mass is explicitly:

$$\begin{aligned}\mathsf{Q}_t^{(\mathbf{r})}(\{\varnothing\} \mid C_t) = \mathfrak{K}_H^Q(\bar{r}^Q)(\varnothing, \{\varnothing\})\mathsf{Q}_B(\{\varnothing\} \mid C_t) \\ + \int_{\mathcal{Z}_H^+} \mathfrak{K}_H^Q(\bar{r}^Q)(z, \{\varnothing\})\, \mathsf{Q}_B(dz \mid C_t).\end{aligned}$$

Thus a nonzero residual may change the no-event probability by moving mass in either direction. Let $\mathrm{lift}_H : \mathcal{E} \times \mathcal{Z}_H^+ \to \mathcal{E}$ be a declared measurable map that aligns a structured event template with the mark and time selected by d_H. Define the no-event-capable structured point summary:

$$\hat{e}_t^H(\mathbf{r}) = \begin{cases} \varnothing, & d_H(\mathsf{Q}_t^{(\mathbf{r})}) = \varnothing, \\ \mathrm{lift}_H\Big(b_t \oplus_E \bar{r}^E, d_H(\mathsf{Q}_t^{(\mathbf{r})})\Big), & d_H(\mathsf{Q}_t^{(\mathbf{r})}) \in \mathcal{Z}_H^+. \end{cases}$$

$$\mathcal{O}_t(\mathbf{r}) = \Big(\mathsf{Q}_t^{(\mathbf{r})}(\cdot \mid C_t), \hat{e}_t^H(\mathbf{r})\Big) \in \mathcal{P}(\mathcal{Z}_H) \times \mathcal{E}_\varnothing.$$

The residual record pairs two independently typed semantics. The law component controls the proper forecast, while the point component controls auxiliary structured fields. The fixed d_H and lift_H keep the final mark and time coherent with the corrected law, but they do not prove that auxiliary fields improved; a joint record must pass forward validation of the complete output bundle. The baseline bundle is exactly $\mathcal{O}_t^B = \mathcal{O}_t(\mathbf{0}_R) = (\mathsf{Q}_B(\cdot \mid C_t), \hat{e}_t^H(\mathbf{0}_R))$. Form the selected candidate $\mathcal{O}_t^{\mathrm{cand}} = \mathcal{O}_t(\mathbf{r}_t^{\mathrm{use}})$, and accept it only from current information:

$$J_t^{\mathrm{pre}} = \mathbf{1}\Big[\mathcal{R}_{\mathrm{pre}}(\mathcal{O}_t^{\mathrm{cand}} \mid C_t; S_{t^-}) \le \eta_{\mathrm{pre}}\Big].$$

The final residual law and bundle are:

$$\mathcal{O}_t^R = \begin{cases} \mathcal{O}_t^{\mathrm{cand}}, & J_t^{\mathrm{pre}} = 1, \\ \mathcal{O}_t^B, & J_t^{\mathrm{pre}} = 0, \end{cases} \qquad \mathsf{Q}_t^R = \begin{cases} \mathsf{Q}_t^{(\mathbf{r}_t^{\mathrm{use}})}, & J_t^{\mathrm{pre}} = 1, \\ \mathsf{Q}_B, & J_t^{\mathrm{pre}} = 0. \end{cases}$$

An implementation may try the next lower-precedence residual after a rejected candidate only when that fallback order and every gate were preregistered. No post-observation quantity may enter this decision.

If an implementation supplies only the point operator $\oplus_E$ and no declared law component and kernel $\mathfrak{K}_H^Q$, it may claim improvement only on point diagnostics, not on the proper forecast score. Conversely, a law-only record may support a proper-score claim but does not claim correction of auxiliary template fields.

Worked toy instantiation. The following finite example is arithmetic scaffolding, not an experimental result. Let $H = 1$ second and order the outcome space as

$$\mathcal{Z}_H = \{z_a, z_b, \varnothing\}, \qquad z_a = (\text{move}, 0.2), \qquad z_b = (\text{stop}, 0.8).$$

For one context C_t, suppose the baseline law is the row vector $\mathbf{q}_B = (0.50, 0.20, 0.30)$.

Take $\mathscr{H} = \mathbb{R}^3$, represent both tagged residual components by diagonal self-adjoint matrices, and use the certified joint cache record

$$r_0^E = r_0^Q = \operatorname{diag}(0.10, -0.05, -0.05), \qquad \mathbf{r}_0 = (r_0^E, r_0^Q, EQ), \qquad \delta_E = \delta_Q = 0.20.$$

The equality of the two matrices is a convenience of this toy, not a semantic identification. Their Frobenius norm is $\sqrt{0.015} < 0.20$, so clipping leaves both unchanged. For completeness, one distributional estimator on this finite space is

$$\rho_H^Q(\mathbf{q}, z) = \operatorname{diag}(\mathbf{1}_z - \mathbf{q}),$$

where $\mathbf{1}_z$ is the one-hot vector for any $z \in \mathcal{Z}_H$, including $\varnothing$. A separately declared point estimator supplies r_0^E. A cache may store projected or averaged outputs such as $\mathbf{r}_0$, together with component-specific estimator identities, horizon, and provenance; the example does not claim that one observation produced either component.

Define

$$\lambda(r^Q) = \min\left(1, \max\left(0, \frac{\langle r^Q, r_0^Q \rangle_F}{\|r_0^Q\|_F^2}\right)\right)$$

and let the full-outcome kernel, in the displayed outcome order, be the row-stochastic matrix

$$K(r^Q) = \begin{pmatrix} 1 & 0 & 0 \\ \lambda(r^Q)/4 & 1 - \lambda(r^Q)/4 & 0 \\ \lambda(r^Q)/6 & 0 & 1 - \lambda(r^Q)/6 \end{pmatrix}.$$

Set $\mathfrak{K}_H^Q(r^Q)(z_i, \{z_j\}) = K(r^Q)_{ij}$. Every row sums to one, all entries are non-negative, and $K(0_Q) = I$. At r_0^Q, the corrected law is

$$\mathbf{q}_R = \mathbf{q}_B K(r_0^Q) = (0.60, 0.15, 0.25).$$

The law correction therefore moves 0.05 probability from z_b and 0.05 from $\varnothing$ to z_a; the no-event branch is operational rather than pinned. Let d_H select a mode under zero-one loss with the displayed order as tie-break. Let the baseline event template be $b_t = e_a \in \mathcal{E}$, whose mark and anchor correspond to z_a, and set $q_E(e_a) = \operatorname{diag}(1, 0, 0)$. Let $\mathcal{Q}_{E,\text{adm}}$ be the diagonal probability simplex, let Π_E be Euclidean projection onto it, and let d_E decode its largest coordinate into the corresponding marked template with the same tie-break. Then $e_a \oplus_E r_0^E = e_a$. With lift_H replacing the template's mark and temporal anchor by the marked decision, the coherent summary is

$$d_H(\mathbf{q}_R) = z_a, \qquad \hat{e}_t^H(\mathbf{r}_0) = \text{lift}_H(e_a, z_a) = e_a.$$

If z_a is observed, logarithmic loss changes from $-\log(0.50) \approx 0.693$ to $-\log(0.60) \approx 0.511$. If $\varnothing$ is observed, it worsens from $-\log(0.30) \approx 1.204$ to $-\log(0.25) \approx 1.386$. This paired calculation shows why a residual needs forward evidence and cannot be certified from one favorable case.

Suppose the exact cache entry records $H_{k_t} = 1$, all other metadata gates pass, and the requested horizon is $H = 1$. Then $J_t^A = 1$ and $\mathbf{r}_t^{\text{use}} = \mathbf{r}_0$. The same entry requested at $H = 0.5$ has $J_t^A = 0$ solely because $H_{k_t} \neq H$, so the expired-horizon correction is not reused.

Finally, let the finite design family built on $\mathcal{S}_{\text{obj}}$ be $\{\Theta_0, \Theta_1\}$, where Θ_0 is baseline-only and Θ_1 includes the certified residual. Take $\epsilon_{AP} = 0.05$, $D_K^{\text{cert},\star}(\Theta_0) = 0.03$, $D_K^{\text{cert},\star}(\Theta_1) = 0.04$, $\epsilon_{\text{prop}} = 0.02$, and suppose the grouped design-sample calculations are

$$
\begin{gathered}
\left(\widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{obj}}}(\Theta_0), \widehat{\mathcal{R}}_{\text{prop}}^{\mathcal{S}_{\text{obj}}}(\Theta_1)\right) = (0.80, 0.72), \\
\left(\widehat{\mathcal{R}}_{\text{pri}}^{\mathcal{S}_{\text{obj}}}(\Theta_0), \widehat{\mathcal{R}}_{\text{pri}}^{\mathcal{S}_{\text{obj}}}(\Theta_1)\right) = (0.30, 0.24), \\
(\mathcal{C}_{\text{rep}}(\Theta_0), \mathcal{C}_{\text{rep}}(\Theta_1)) = (0, 0.01).
\end{gathered}
$$

with $\text{UCB}[0.72 - 0.80] = -0.02 \leq \epsilon_{\text{prop}}$ and $\lambda_{\text{rep}} = 1$. Both designs remain feasible, while their empirical composite values are 0.30 and 0.25, so the deterministic operational rule selects $\widehat{\Theta}_\Gamma = \Theta_1$. An untouched $\mathcal{S}_{\text{conf}}$ may then confirm or reject the frozen claim, but it cannot alter the candidate family or the selected residual. These stipulated values demonstrate how to execute the contracts; they are not measurements of EventFrame performance.

A bounded hash table can provide expected $O(1)$ lookup after the bounded key has been constructed. The epoch and margin are constant-size certificate checks; graph traversal and compatibility estimation remain off the hot path. Key construction, hashing, collision handling, synchronization, and eviction remain separate costs. The active epoch v_t must increase whenever a dependent comparison map, edge set, threshold, or simultaneous defect bound changes. Local epochs and a reverse dependency index permit affected entries to be invalidated without globally flushing unrelated abstractions. A predictive sheaf snap is built against a shadow graph version and published atomically with its affected abstraction-key map and epoch map. A reader uses one immutable graph-key-epoch snapshot for the entire prediction; it must never combine a new graph or abstraction key with old certificates. Entries invalidated by the publication fall back to the baseline or another currently certified cache path until recertified, and rollback republishes the previous complete structure with a new monotone publication version rather than reusing an old epoch identifier.

After observation, evaluate the particular residual candidate stored for key k, either on a deployed trial or in shadow mode. Set $I_{t,k} = 1$ when

$$
\mathcal{A}_{\text{post}}(\mathcal{O}_t^B, Z_{t+1}) - \mathcal{A}_{\text{post}}(\mathcal{O}_t(\mathbf{r}_k), Z_{t+1}) \geq \delta_A,
$$

and set $I_{t,k} = 0$ otherwise, where $\delta_A > 0$. A fallback residual belonging to another key may not update this candidate's confidence. Under an explicitly stationary, conditionally independent Bernoulli model for evaluated trials assigned to key k, with prior $\text{Beta}(a_0, b_0)$ and $a_0, b_0 > 0$, the posterior mean is:

$$c_{k,t^-} = \frac{a_0 + \sum_{u \in \mathcal{T}^{\text{cache}}_{k,t^-}} I_{u,k}}{a_0 + b_0 + |\mathcal{T}^{\text{cache}}_{k,t^-}|}.$$

Here $\mathcal{T}^{\text{cache}}_{k,t^-}$ contains only trials for that exact candidate whose outcome availability time satisfies $a(Z_{u+1}) \le t$. The Beta update is justified only for conditionally independent episode-level units. Overlapping windows from the same trajectory must be clustered or replaced by a declared effective-support calculation; they may not be counted as independent trials. This posterior mean is not automatically calibrated; calibration is tested on forward-held-out trials. Repeated monitoring uses a confidence sequence, alpha-spending rule, or fixed preregistered review times rather than repeatedly applying a fixed-sample interval. For drift, the implementation may use explicitly time-decayed counts, but must report the decay schedule and effective sample size. Low confidence, insufficient support, excessive pre-risk, or worsened post-loss routes the case to slow-path review.

The runtime packet uses a separate typed composition operator. Let:

$$X_t = \chi(C_t, \mathcal{M}_t, G_t, \sigma_t) \in \mathcal{X}_{\text{ctx}},$$

and define the packet space:

$$\mathcal{Y}_{\text{pkt}} = \mathcal{N}_{\text{mem}} \times \mathcal{E}_{\text{graph}} \times \mathcal{L}_{\text{lane}} \times \mathcal{C}_{\text{compact}} \times \mathcal{M}_{\text{mode}} \times \mathcal{U}_{\text{control}}.$$

Choose a normed finite-dimensional packet representation $\mathcal{V}_Y$, a non-empty closed admissible subset $\mathcal{V}_{Y,\text{adm}}$, and maps:

$$q_Y : \mathcal{Y}_{\text{pkt}} \to \mathcal{V}_Y, \qquad d_Y : \mathcal{V}_{Y,\text{adm}} \to \mathcal{Y}_{\text{pkt}},$$

with a deterministic projection selection $\Pi_Y(v) \in \arg\min_{u \in \mathcal{V}_{Y,\text{adm}}} \|u - v\|_Y$ and clipping radius $\delta_Y > 0$. Define:

$$y \oplus_Y r = \begin{cases} y, & r = 0, \\ d_Y\big(\Pi_Y\big(q_Y(y) + \text{clip}_{\delta_Y}(r)\big)\big), & r \neq 0. \end{cases}$$

The baseline and residual now have compatible types:

$$B_Y : \mathcal{X}_{\text{ctx}} \to \mathcal{Y}_{\text{pkt}}, \qquad R_Y : \mathcal{X}_{\text{ctx}} \to \mathcal{V}_Y,$$

and the packet prediction is:

$$\widehat{\mathbf{y}}_{t+1} = B_Y(X_t) \oplus_Y R_Y(X_t).$$

Its components are top memory nodes, top graph edges, retrieval lane, compaction risk, response mode, and an optional control branch. Discrete components may be encoded as logits with validity masks; the decoder must specify tie-breaking and null actions.

After execution, let $\mathbf{y}^{\star}_{t+1}$ be the audited packet target and let $\mathcal{L}_{\mathrm{pkt}}(\widehat{\mathbf{y}}, \mathbf{y}^{\star}) \in [0,1]$ be a declared weighted component loss. Packet residual utility is the observed improvement:

$$I_t^Y = \mathbf{1}\left[\mathcal{L}_{\mathrm{pkt}}(B_Y(X_t) \oplus_Y R_Y(X_t), \mathbf{y}^{\star}_{t+1}) + \delta_{\mathrm{pkt}} \le \mathcal{L}_{\mathrm{pkt}}(B_Y(X_t), \mathbf{y}^{\star}_{t+1})\right].$$

Require $\delta_{\mathrm{pkt}} > 0$, so mere ties do not count as evidence that a packet residual improved utility.

Confidence is updated from the corresponding success/failure counts, as above. If $\mathcal{P}_t = \{(p_m, w_m)\}_{m=1}^{M}$ is a non-empty candidate set with $w_m \ge 0$, $\sum_m w_m > 0$, and $\lambda_P \ge 0$, let $\ell_t^{(m)} \in [0,1]$ be the declared post-observation loss appropriate to candidate p_m, such as packet loss for packet candidates or event action for event candidates. An explicitly heuristic exponential-weights update is:

$$w_m^{\mathrm{new}} = \frac{w_m \exp(-\lambda_P \ell_t^{(m)})}{\sum_{j=1}^{M} w_j \exp(-\lambda_P \ell_t^{(j)})}.$$

This is not a Bayesian particle filter unless $\ell_t^{(m)}$ is a negative log-likelihood with the required probabilistic model. Pruning or resampling must monitor effective sample size to avoid premature collapse.

The main failure modes are cache pollution, overcorrection, stale residuals, false similarity, invalid decoding, and packet-component incompatibility. Every implementation must report cache support, age, pre-risk, realized improvement, fallback frequency, and decoder failures.

5. Memory Model

EventFrame uses memory for two different purposes: recalling prior events and reusing prior corrections. These purposes should not be collapsed. Episodic memory stores cases. A residual cache stores adjustments to a baseline prediction. Both may support prediction, but they answer different questions.

An episodic key-value cache can be written:

$$\mathcal{C}_E = \{(u_i, v_i, s_i)\}_{i=1}^{M},$$

where u_i is a retrieval key, v_i is an event frame, trajectory segment, or summary, and s_i is metadata. Given a context C_t, an episodic lookup retrieves prior cases that resemble the current situation. The operational use is case recall: retrieve examples that may inform the baseline model, explain the current state, or provide analogies for review.

A residual cache is different:

$$\mathcal{C}_R = \{(\kappa_i, r_i, s_i)\}_{i=1}^{N}.$$

Here r_i is not a prior event. It is a correction to a prior baseline prediction. The operational use is correction reuse: if the current context resembles a past context where the baseline missed in a known direction, apply the cached residual through the typed operator $\oplus_E$.

The conceptual distinction is important. Episodic memory says, "something like this happened before." Residual memory says, "the predictor made this kind of mistake before." A system can have useful episodic recall but poor residual reuse if prior cases are similar but their prediction errors differ. Conversely, a residual may be reusable even when the full episode is not otherwise relevant.

Prediction combines the two memories by priority rather than by collapse. A reference flow is:

1. Compute the baseline $b_t = B(C_t)$.
2. Try action-residual lookup in $\mathcal{C}_A$.
3. If the action residual record is valid under confidence, age, horizon, provenance, and pre-risk checks, apply its point and law components to their separately declared outputs.
4. If confidence is insufficient, try residual lookup in $\mathcal{C}_R$.
5. If residual confidence is still insufficient, retrieve episodic cases from $\mathcal{C}_E$ and use them to refine the baseline, explain uncertainty, or schedule slow-path review.
6. After observation, update episodic memory, residual confidence, and any action-residual entry that was used or falsified.

This flow keeps the low-latency path cheap while preserving a fallback to richer case evidence. Residual memory can answer quickly when the current situation matches a known error pattern. Episodic memory becomes more important when the residual cache is missing, low-confidence, stale, or contradicted by recent outcomes.

Similarity lookup requires declared key functions and distances. For episodic memory, the key function may emphasize entities, action types, and temporal neighborhoods. For residual memory, the key should emphasize features that predict baseline error. These are not necessarily the same. For example, two events may share an action type but differ in timing dynamics; they may be episodically similar while producing different residuals.

Consolidation is the process of updating memory after observation. A conservative consolidation step should:

1. Record the observed event e_{t+1} with provenance and confidence.
2. Compute proper predictive loss and the event-aware timing diagnostic.
3. Estimate whether the baseline error is systematic enough to store as a residual.
4. Update or decay cache entries based on age, confidence, and repeated utility.
5. Preserve at least one traceability frame and the coverage-aware context audit set required by Section 7.
6. Mark low-confidence entries so they cannot dominate future predictions.

Cache pollution is the main risk. If every error becomes a residual, the cache may memorize noise. If keys are too broad, residuals are applied in inappropriate contexts. If keys are too narrow, useful residuals are never reused. The cache should therefore track hit rate, post-correction loss, and whether retrieved residuals improve over the baseline.

Fast-path memory use should be cheap. A practical implementation may use approximate nearest-neighbor lookup, hashed keys, or bounded-size caches. The paper treats constant-time lookup as an approximation, not as a guarantee. Slow-path memory refinement may be more expensive because it runs after the initial prediction, when latency pressure is lower.

Representative preservation is a memory responsibility. A single traceability frame prevents a group from becoming an empty label, but boundary detection requires the context audit set, its associated anchor frames, coverage metadata, and sampling history. If these are discarded, the runtime must mark the group unaudited rather than infer stability from one example.

Selective Bayesian Update Frontier

EventFrame may attach a bounded Bayesian belief state to an event bucket, residual family, latent regime, or declared hypothesis family. It does not update every stored belief after every frame. Instead, vector retrieval and graph locality propose a finite update frontier. Let $\mathcal{R}_t^{\mathrm{vec}}$ be at most k_v candidates returned by the frozen vector-retrieval rule. Let $\mathcal{N}_t^{\mathrm{sh}}$ be the bounded neighborhood returned by the abstraction compatibility graph. This is a sheaf-inspired neighborhood, not a sheaf-theoretic neighborhood unless the required restriction identity and composition laws have actually been instantiated.

If an explicit SCM $\mathfrak{M}$ exists, let v_t^E be the graph node associated with the current context and use its declared parents and children. A child here is an outgoing relationship already present in the as-of graph, not a future realized event. Without an identified SCM, the corresponding predictive-dependency neighbors may be used but may not be called causal. The candidate frontier is

$$\mathcal{N}_t^B = \mathcal{R}_t^{\mathrm{vec}} \cup \mathcal{N}_t^{\mathrm{sh}} \cup \begin{cases} \mathrm{Pa}_{\mathfrak{M}}(v_t^E) \cup \mathrm{Ch}_{\mathfrak{M}}(v_t^E), & \mathfrak{M} \text{ is available}, \\ \mathcal{N}_t^{\mathrm{pred}}, & \text{otherwise}. \end{cases}$$

Every set is constructed from the as-of snapshot and has a predeclared cardinality or degree cap. For an evidence-bearing event $e \in \mathcal{N}_t^B$, define four measurable scores in $[0,1]$: vector relevance $v_t^B(e)$, sheaf-inspired neighbor compatibility $n_t^B(e)$, novelty $u_t^B(e)$, and source independence $s_t^B(e)$. Freeze non-negative weights satisfying $\alpha_B + \beta_B + \gamma_B + \delta_B = 1$, and set

$$A_t^B(e) = \alpha_B v_t^B(e) + \beta_B n_t^B(e) + \gamma_B u_t^B(e) + \delta_B s_t^B(e).$$

Let $c_t^B(e) \in [0,1]$ be structural criticality available before the downstream target whose performance will be evaluated. For fixed $0 \le \tau_{\min} \le \tau_{\max} \le 1$, define the lower threshold for critical neighbors by

$$\tau_t^B(e) = \min\bigl(\tau_{\max}, \max\bigl(\tau_{\min}, \tau_0 - \lambda_{\mathrm{crit}} c_t^B(e)\bigr)\bigr), \qquad J_t^{\mathrm{act}}(e) = \mathbf{1}[A_t^B(e) \ge \tau_t^B(e)].$$

All scoring maps, normalizations, weights, thresholds, tie-breaks, and source-dependence estimators are part of Λ_{eval}. A score may use a newly arrived frame once that frame is available, but it may not use a later target outcome, posterior audit, or graph revision. Activation controls expenditure; it does not establish that candidates are safe to pool.

Anti-Pigeon controls posterior granularity. For a candidate bucket K, let v_K^B be the abstraction epoch under which its posterior-sharing certificate was produced. Sharing is permitted only when

$$J_{K,t}^{\text{share}} = \mathbf{1}\Big\{ D_K^{\text{cert},\star} \le \epsilon_{B,\text{share}}, \quad n_K^{\text{eff}} \ge n_{B,\min}, \quad v_K^B = v_t, \quad H_K = H, \quad s_K^B \text{ is valid} \Big\}.$$

The certificate concerns externally evaluated downstream target-law disagreement, not agreement among the candidate model's own posteriors. The fast path checks a materialized certificate; it does not recompute $D_K^{\text{cert},\star}$. Active events in a certified bucket may update one shared posterior. If the certificate fails or is unavailable, each event retains or receives a separate posterior and the case may be routed to slow-path split review. Unrelated events are ignored by the production update except for the audit and changepoint mechanisms below.

Let $(\Theta_K, \mathscr{A}_{\Theta_K})$ be a declared parameter space and let $q_{K,t^-} \in \mathcal{P}(\Theta_K)$ be the cached prior available before the update. Let $\xi_t(e)$ be the evidence packet extracted from an available event and its currently available labels. For an activated, sharing-approved evidence set $\mathcal{E}_{K,t}^{\text{act}}$, an ordinary Bayesian update is

$$q_{K,t}^{+}(d\theta) = \frac{L_K^{\text{sel}}(\mathcal{E}_{K,t}^{\text{act}} \mid \theta, \mathfrak{h}_t) q_{K,t^-}(d\theta)}{\int_{\Theta_K} L_K^{\text{sel}}(\mathcal{E}_{K,t}^{\text{act}} \mid \vartheta, \mathfrak{h}_t) q_{K,t^-}(d\vartheta)},$$

provided the denominator is finite and strictly positive. Because novelty or compatibility may depend on the arrived event, activation is generally informative. For one evidence packet ξ, the selection-conditioned likelihood is

$$L_K^{\text{sel}}(\xi \mid \theta, \mathfrak{h}_t, J^{\text{act}} = 1) = \frac{P_\theta(J^{\text{act}} = 1 \mid \xi, \mathfrak{h}_t) p_\theta(\xi \mid \mathfrak{h}_t)}{P_\theta(J^{\text{act}} = 1 \mid \mathfrak{h}_t)},$$

on the domain where the denominator is positive. For a jointly activated evidence set, the contract must model the joint activation probability; multiplying one-event selection corrections is valid only under a declared conditional factorization. Selection may be ignored only under a stated conditional-ignorability result, for example when activation depends exclusively on already conditioned-on pre-evidence variables. If the selection probability cannot be modeled, the result is called an activation-conditioned working posterior, not a calibrated posterior for the full event stream, and must be tested against the independent audit stream.

Likewise, a product of conditionally independent likelihoods is ordinary Bayes only when the declared source model justifies that factorization. Tempering correlated-source contributions,

$$q_{K,t}^{+}(d\theta) \propto q_{K,t^-}(d\theta) \prod_{e \in \mathcal{E}_{K,t}^{\text{act}}} p_\theta(\xi_t(e) \mid \mathfrak{h}_t)^{\omega_t(e)}, \qquad 0 \le \omega_t(e) \le 1,$$

defines a generalized or power posterior unless it is derived from a joint generative model. Source-independence scoring therefore cannot by itself justify multiplying evidence as if it were independent.

For cheap regime monitoring, a bucket may maintain a bounded approximation to a Bayesian online changepoint run-length posterior [17,18]. Let $R_{K,t} \in \mathbb{N}_0$ be run length and define

$$J_{K,t}^{\mathrm{cp}} = \mathbf{1}[P(R_{K,t} = 0 \mid \mathfrak{h}_t) \geq \gamma_{\mathrm{cp}}] .$$

When this indicator fires, the runtime invalidates the affected shared posterior and residual certificate, increments the local epoch, expands the review frontier, and routes recalibration to the slow path. A monitor fed only activated evidence detects changes in the selected process. It supports a full-stream regime claim only when its transition and observation model includes the selection mechanism or when the independent audit stream is incorporated with its sampling design. Exact classical run-length support can grow with the stream; a constant-memory or constant-time claim therefore requires a declared cap, pruning rule, or finite sufficient-statistic approximation and must report its approximation error.

Selective retrieval can become self-confirming by never revisiting what it has learned to ignore. EventFrame therefore reserves a predeclared audit probability $\pi_{\mathrm{audit}} > 0$. Conditional on the inactive candidate set and independently of activation-score magnitude, draw

$$J_t^{\mathrm{audit}}(e) \sim \mathrm{Bernoulli}(\pi_{\mathrm{audit}}).$$

If the accepted audit sample exceeds a fixed budget B_{audit}, a frozen uniform reservoir subsamples it and records every final inclusion probability. Audit estimators use the corresponding design weights; an unweighted capped convenience sample cannot support the omission certificate. Audited inactive events are evaluated in shadow mode against an expanded update. Define the omitted-influence statistic

$$U_t^{\mathrm{omit}} = \mathrm{UCB}_{\mathrm{sim}}\left[D_B\left(\mathsf{Q}_t^{\mathrm{local}}, \mathsf{Q}_t^{\mathrm{expanded}}\right)\right] .$$

Local updating remains certified only while $U_t^{\mathrm{omit}} \leq \epsilon_{B,\mathrm{omit}}$. The simultaneous procedure covers every inspected bucket, expansion, and repeated audit decision. Every production or shadow decision records $(A_t^B, \tau_t^B, J_t^{\mathrm{act}}, J_{K,t}^{\mathrm{share}}, J_t^{\mathrm{audit}}, v_t, H, s_t^{\mathrm{prov}})$, together with audit inclusion probability, so calibration can be reconstructed under as-of replay.

Posterior, posterior key, dependent residual certificate, and epoch publish atomically. Prediction readers observe one complete old or new version, never a mixed state. Posterior storage has a declared capacity and deterministic eviction rule. Eviction removes fast-path reuse eligibility but preserves immutable provenance required by later audits.

Streaming variational Bayes motivates incremental and asynchronous posterior approximation [14]. Streaming variational Monte Carlo and online variational sequential Monte Carlo provide richer state-space and particle-based alternatives [15,16], but their constant-per-sample or online properties do not make their particle count, parameter dimension, optimization, or hardware cost free. Pattern Markov Chains are relevant only for declared event-pattern completion forecasts, not as a universal next-event Bayesian model [19]. Work on out-of-distribution sequential event prediction motivates latent-context and shift-aware evaluation [20], but EventFrame does not inherit its causal interpretation without the corresponding identification assumptions.

The memory model supports the overall EventFrame loop. Episodic memory helps interpret and compare cases. Residual memory corrects recurring transition errors. The selective Bayesian frontier updates bounded cached beliefs, while Anti-Pigeon decides which evidence may share one posterior. Slow-path consolidation, changepoint review, and independent audits keep all three memories from turning into overconfident filtered history. The next section uses perturbation rather than recall to discover which event properties are stable under prediction.

6. Fuzzing and Invariants

Property fuzzing tests model sensitivity: perturb a selected event field, rerun prediction, and measure the change in a declared output. It does not by itself establish how the real world would respond to an intervention.

Let ϕ_i be an event property. A validity-constrained fuzzing operator is:

$$\mathcal{F}_{i,\epsilon} : \mathcal{E} \rightharpoonup \mathcal{E},$$

where the partial arrow records that some perturbations are invalid. At context position or subset r:

$$\mathcal{F}_{i,\epsilon}^{(r)} : \mathcal{E}^k \rightharpoonup \mathcal{E}^k.$$

Let $\mathcal{O}_\theta(C) = (\mathsf{Q}_\theta(\cdot \mid C), \hat{e}_\theta^H(C))$ be the typed predictor output, including its coherent no-event-capable point summary. For a declared output functional g on that bundle and distance d_g, model sensitivity is:

$$\Delta_g^{\text{model}} = d_g\Big(g(\mathcal{O}_\theta(C_t)), g(\mathcal{O}_\theta(\mathcal{F}_{i,\epsilon}^{(r)}(C_t)))\Big).$$

The validation law $\mathcal{V}_i$ must be supported only on triples (C_t, ϵ, r) for which the partial perturbation is defined. The field is empirically stable over that declared valid family when:

$$\Pr_{(C_t,\epsilon,r)\sim\mathcal{V}_i}\Big(\Delta_g^{\text{model}} \le \eta_g\Big) \ge 1 - \alpha_g,$$

with a one-sided lower confidence bound for this probability at least $1 - \alpha_g$. A point estimate or a two-sided interval that crosses the threshold does not establish stability. The reporting score

$$S_g = \min\left(1, \frac{\Delta_g^{\text{model}}}{\eta_g}\right)$$

requires $\eta_g > 0$. Thresholds are selected from measurement resolution, operational decision tolerance, and held-out calibration; fixed fractions such as $0.05H$ are examples only and must not be presented as universal constants.

For 5W1H review, let $\psi_j^{\text{role}}(e)$ denote the component assigned to role $j \in \{W, A, T, L, M, H\}$. The average sensitivity of field ϕ_i to target property g is:

$$I_{i \to g}^{\text{model}} = \mathbb{E}_{(C_t, \epsilon, r) \sim \mathcal{V}_i} \left[\Delta_g^{\text{model}} \right].$$

This quantity may nominate a field for retain, migrate, duplicate, split, or uncertain status. It says that the current predictor uses the field; it does not prove that the field is a cause, that the assigned semantic explanation is true, or that changing the field in the world would change the target.

An operational protocol is:

1. Select contexts, target property, field, perturbation family, and validity constraints.
2. Separate observed contexts from synthetic perturbations.
3. Run original and perturbed predictions.
4. Estimate sensitivity, uncertainty, and boundary regions on held-out contexts.
5. Check whether the result survives alternative plausible perturbation families.
6. Use the result as a review signal, not an automatic ontology rewrite.

Synthetic frames are never inserted into episodic memory as observations. They may be stored in a separate audit log with their generating operator and validity assumptions.

Graph perturbation follows the same rule. Let $G_t = (V_t, R_t)$ be a time-unrolled predictive graph and let:

$$G_t' = \mathcal{I}_{v,\epsilon}^{\text{model}}(G_t).$$

The resulting predictor sensitivity is:

$$\Delta_Y^{\text{model}} = D_Y^{\text{law}} \left(\mathsf{Q}_\theta^Y(\cdot \mid G_t'), \mathsf{Q}_\theta^Y(\cdot \mid G_t) \right),$$

where Q_θ^Y is the declared predictive marginal for target Y. This may update predictive-dependency confidence, residual keys, or abstraction review priorities. It must not update causal-edge confidence merely because the predictor changed.

When an explicit structural causal model $\mathfrak{M} = (U, V, F, P_U)$ exists and an intervention target is well-defined, a separate causal analysis may compute:

$$\Delta_Y^{\text{causal}} = D_Y^{\text{law}} \left(P_{\mathfrak{M}}(Y \mid do(V_j = v')), P_{\text{ref}}(Y) \right).$$

The reference law P_{ref} must be declared, and this distance is an effect magnitude rather than a signed effect. Identification assumptions, manipulated variables, confounder controls, and transport assumptions must be stated. Randomized or otherwise identified intervention evidence may update causal-edge confidence; input fuzzing alone may not [5].

The slow path begins only after a realized post-observation loss is available:

1. Observe $\mathcal{A}_{\text{post}} > \eta_{\text{post}}$ or repeated packet failure.
2. Select candidate fields, nodes, or edges from residual and uncertainty evidence.
3. Run validity-constrained model perturbations.
4. If an SCM and identification strategy exist, run the corresponding causal analysis separately.
5. Update cache keys, predictive edges, or abstraction markers only after repeated held-out improvement.

For a candidate ontology change from state s to s', use an independent paired forward-validation set $\mathcal{V}_{\text{rev}} = \{(C_t, Z_{t+1})\}_{t=1}^{n}$. Replay each case from S_{t^-}, include it only when $a(Z_{t+1})$ is inside the validation availability window, and group inference by independent trajectory or entity. Define per-case composite improvement:

$$\Delta_t^{s \to s'} = \mathcal{A}_{\text{post}}(\mathcal{O}^{s}(C_t), Z_{t+1}) - \mathcal{A}_{\text{post}}(\mathcal{O}^{s'}(C_t), Z_{t+1}).$$

and the paired proper-score degradation:

$$G_{t,\text{prop}}^{s \to s'} = S_{\text{prop}}(\mathsf{Q}^{s'}(\cdot \mid C_t; S_{t^-}), Z_{t+1}) - S_{\text{prop}}(\mathsf{Q}^{s}(\cdot \mid C_t; S_{t^-}), Z_{t+1}).$$

Promotion requires all of the following preregistered conditions:

$$n \geq n_{\min}^{\text{rev}}, \qquad \text{LCB}_{\text{paired}}\left[\frac{1}{n}\sum_{t=1}^{n} \Delta_t^{s \to s'}\right] \geq \delta_{\text{rev}} > 0,$$

$$\text{UCB}\left[\frac{1}{n}\sum_{t=1}^{n} \mathbf{1}\{\Delta_t^{s \to s'} < -\delta_{\text{harm}}\}\right] \leq \beta_{\text{harm}}.$$

It additionally requires proper-score non-inferiority:

$$\text{UCB}_{\text{paired}}\left[\frac{1}{n}\sum_{t=1}^{n} G_{t,\text{prop}}^{s \to s'}\right] \leq \epsilon_{\text{prop}}^{\text{rev}}, \qquad \epsilon_{\text{prop}}^{\text{rev}} \geq 0.$$

Here $\delta_{\text{harm}} \geq 0$ and $\beta_{\text{harm}} \in [0, 1]$ are fixed before evaluation.

Thus average composite improvement cannot hide either an uncontrolled rate of material regressions or degraded probabilistic calibration. The confidence construction must account for every adaptively compared candidate state. If promotion is monitored repeatedly, use a confidence sequence, alpha spending, or preregistered review times. All learned preprocessing, perturbation selection, and priority rules are fitted before the validation cutoff. The evaluation contexts must not be the same or temporally overlapping examples used to propose the change. Before validation, the field remains provisional. Previous assignments and provenance are retained so the change can be audited or reversed.

An EventFrame invariant is therefore conditional: stable under this valid perturbation family, for this predictor and target, in this data regime, within this threshold and confidence level. Failure modes include invalid perturbations, off-manifold inputs, hidden confounding, adaptive reuse of the validation set, and thresholds below measurement noise.

7. Lumpability and Abstraction

Abstraction is useful only when it preserves the transition behavior required by the declared target. Let:

$$\pi : \mathcal{E} \to \mathcal{S}_{\text{abs}}$$

map detailed events to abstract states, and extend it componentwise to contexts as $\pi^k(C_t)$.

Let $\mathfrak{C}_{\text{adm}} \subseteq \mathcal{E}^k$ be the declared admissible context domain from Section 3. The target Y, target law $P_\star$, divergence, and admissible context domain are fixed by the evaluation contract before π is selected. An aggregate conditional law is not by itself a lumpability test because it averages over hidden detailed states inside a bucket. Instead, define the external predictive lumpability defect:

$$\varepsilon^\star_{\text{lump}}(\pi) = \sup_{C, C' \in \mathfrak{C}_{\text{adm}} : \, h_\pi(C) = h_\pi(C')} D\big(P_\star(Y \mid C), P_\star(Y \mid C')\big) .$$

The abstraction is ϵ_L-predictively lumpable for the target when:

$$\varepsilon^\star_{\text{lump}}(\pi) \le \epsilon_L .$$

This pairwise condition prevents an aggregate conditional distribution from hiding incompatible microstate transitions. It adapts classical and near-lumpability to finite-context prediction rather than claiming a new Markov-chain theorem [3,4]. In finite data, the supremum is estimated with confidence bounds over observed or generated context pairs; passing the estimate is evidence, not proof about unseen contexts.

Operationally:

1. Freeze the target, target law, divergence D, tolerance ϵ_L, and evaluation protocol; then choose π.
2. Form detailed context pairs that map to the same operational key h_π.
3. Compare their fixed-target future distributions.
4. Report the maximum estimated divergence with uncertainty and minimum bucket support.
5. Accept the abstraction only when held-out predictive degradation and the upper confidence bound remain below threshold.

Confluence applies the same requirement to merged event streams. Divergence rejects a merge when a small valid perturbation produces target-distinct future distributions. These statements concern predictive equivalence unless a separate causal model supports intervention claims.

Every non-empty bucket $K \subseteq \mathcal{E}$ retains at least one concrete frame $\bar{e}_K \in K$ for traceability, but one frame is not sufficient to characterize a heterogeneous bucket. Let $\mathrm{anc}(C) = e_t$ denote the terminal or anchor frame of context $C = e_{t-k+1:t}$, and define the context family represented by K:

$$\mathfrak{C}_K = \{C \in \mathfrak{C}_{\mathrm{adm}} : \mathrm{anc}(C) \in K\}.$$

When $\mathfrak{C}_K \neq \varnothing$, call K active and maintain a non-empty context audit set $\mathcal{R}_C(K) \subseteq \mathfrak{C}_K$. If no context has yet been assigned to the bucket, retain $\bar{e}_K$ for traceability but mark the bucket inactive and unaudited; no future-diameter or admissibility claim is made for it. With a declared context metric d_C, a representational coverage rule for an auditable bucket may be:

$$\sup_{C \in \mathfrak{C}_K} \min_{R \in \mathcal{R}_C(K)} d_C(C, R) \le \delta_K.$$

The set should include contexts for a medoid or high-confidence anchor, boundary examples, high-uncertainty examples, and a reservoir sample when the bucket is large. Its associated anchor frames preserve concrete traceability. If compression prevents this coverage estimate, the system cannot claim that the bucket has been audited.

Anti-Pigeon is the split-side guard against invalid abstraction and stale predictive habit. The name denotes anti-pigeonholing: events may share a bucket only while their target futures remain sufficiently similar.

For each bucket K and contexts $C, C' \in \mathfrak{C}_K$, define the external target-law disagreement:

$$D^{K,\star}_{C,C'} = D\big(P_\star(Y \mid C), P_\star(Y \mid C')\big)\,,$$

and the theoretical future-diameter:

$$D^\star_K(\pi) = \sup_{C, C' \in \mathfrak{C}_K} D^{K,\star}_{C,C'}.$$

The bucket is admissible only when:

$$D^\star_K(\pi) \le \epsilon_{AP}.$$

Separately define the model-forecast diameter

$$D^{\mathrm{mdl}}_K(\Theta_\Gamma) = \sup_{C, C' \in \mathfrak{C}_K} D\big(\mathsf{Q}^Y_{\Theta_\Gamma}(\cdot \mid C; S_{t^-}), \mathsf{Q}^Y_{\Theta_\Gamma}(\cdot \mid C'; S_{t^-})\big)\,.$$

This model diameter detects internal inconsistency and drift, but it cannot certify the abstraction: a predictor that emits the same wrong law everywhere has zero model diameter while the external future-diameter may be large.

Define the true restricted audit diameter and its estimator by:

$$D_K^{\text{audit},\star} = \max_{R,R' \in \mathcal{R}_C(K)} D_{R,R'}^{K,\star}, \qquad \widehat{D}_K^{\star} = \max_{R,R' \in \mathcal{R}_C(K)} \widehat{D}_{R,R'}^{K,\star}$$

where $\widehat{D}_{R,R'}^{K,\star}$ estimates target-law disagreement from observed outcomes without using the candidate forecast as ground truth. The audit reports $\widehat{D}_K^{\star}$, coverage, and statistical uncertainty. The deterministic relation is $D_K^{\text{audit},\star} \leq D_K^{\star}$; no sample-wise ordering between $\widehat{D}_K^{\star}$ and $D_K^{\star}$ is asserted. A statistically significant large pairwise divergence is evidence to split or mark the bucket. Representational coverage alone does not make a small estimate a certificate of unseen future behavior.

To obtain a certified upper bound from a non-exhaustive audit, require that D obey the triangle inequality and verify a continuity bound for the forecast map on $\mathfrak{C}_K$. Let $\overline{L}_K^{\text{cert}}$ be either a deterministic uniform bound established analytically or a simultaneous upper confidence bound produced by a predeclared procedure. It must satisfy, at the certificate's stated confidence level:

$$D(P_\star(Y \mid C), P_\star(Y \mid R)) \leq \overline{L}_K^{\text{cert}} d_C(C, R) \qquad \text{for all } C, R \in \mathfrak{C}_K,$$

then the coverage rule implies:

$$D_K^{\star}(\pi) \leq D_K^{\text{audit},\star} + 2\overline{L}_K^{\text{cert}} \delta_K.$$

With statistical estimation, a simultaneous upper confidence certificate is:

$$D_K^{\text{cert},\star} = \max_{R,R' \in \mathcal{R}_C(K)} \text{UCB}_{\text{sim}}[D_{R,R'}^{K,\star}] + 2\overline{L}_K^{\text{cert}} \delta_K.$$

The confidence procedure must jointly cover all adaptively selected audit pairs used by the maximum and any data-estimated continuity bound. A plug-in estimate of $\overline{L}_K$ without uncertainty coverage is not a certificate. If the audit is exhaustive, the coverage term vanishes. If neither exhaustive coverage nor a verified continuity bound is available, the audit supports only an observed-sample claim and cannot certify $D_K^{\star} \leq \epsilon_{AP}$.

Observed operating regimes use a distinct symbol $\zeta_t \in \mathcal{Z}_{\text{reg}}$. On the common-support domain $\mathfrak{C}_{a,b} = \operatorname{supp}(C \mid \zeta_a) \cap \operatorname{supp}(C \mid \zeta_b)$, regime-conditioned predictive divergence is:

$$D_{i,a,b}^{\text{reg}} = D(P_\star(Y \mid C_i, \zeta_a), P_\star(Y \mid C_i, \zeta_b)).$$

This quantity is evaluated only for $C_i \in \mathfrak{C}_{a,b}$. Outside common support it requires a declared overlap and transport model; otherwise it is unidentified and no comparison is reported. The evaluation contract freezes a held-out review window W_{reg}, the minimum number m_{reg} of multiplicity-adjusted exceedances of $\epsilon_{AP}^{\text{reg}}$, and the resulting action before candidate inspection. Meeting that rule is evidence that a shared predictive bucket is stale; the predeclared action may split by regime, condition the cache key on ζ, decay the residual, or mark the abstraction as divergence-sensitive. Post hoc changes to the window, repetition count, threshold, or action invalidate the claim. This adaptation problem is related to concept-drift detection and response [12]. The conditional difference supports predictive adaptation; it is not evidence that ζ is causal unless intervention or identification assumptions establish that fact.

A split operator returns $\{K_1, \ldots, K_m\}$ such that every non-empty active child has sufficient effective support and either exhaustive verification or $D^{\mathrm{cert},\star}_{K_j} \le \epsilon_{AP}$. Singleton buckets always satisfy an empirical pairwise bound, so representation cost, minimum support, untouched confirmation performance, and coverage of future contexts are required to prevent trivial memorization.

Merge and split thresholds should use hysteresis, for example $\epsilon_{\mathrm{merge}} < \epsilon_{AP}$, and changes should be accepted only after a minimum held-out improvement. Abstraction quality reports memory and latency gains alongside predictive degradation, subgroup errors, audit coverage, and split/merge churn.

EventFrame can extend this bucket-local test to a network of heterogeneous abstractions. Let:

$$\mathcal{G}^A_t = (V^A_t, E^A_t)$$

be an abstraction compatibility graph. A node may represent an event group, temporal resolution, sensor, local predictor, or agent. Node i produces a predictive law:

$$\mathsf{Q}_i(\cdot \mid C_t) \in \mathcal{P}(\mathcal{Y}_i).$$

For an edge $e = \{i, j\}$, choose a common measurable comparison space $\mathcal{Y}_e$ and measurable maps $g_{ie} : \mathcal{Y}_i \to \mathcal{Y}_e$ and $g_{je} : \mathcal{Y}_j \to \mathcal{Y}_e$. Their pushforward restrictions are:

$$\mathsf{r}_{ie}\mathsf{Q}_i = (g_{ie})_*\mathsf{Q}_i, \qquad \mathsf{r}_{je}\mathsf{Q}_j = (g_{je})_*\mathsf{Q}_j.$$

Given a declared divergence D_e, the edge compatibility defect is:

$$\delta_e(\mathsf{Q}) = D_e(\mathsf{r}_{ie}\mathsf{Q}_i, \mathsf{r}_{je}\mathsf{Q}_j)\,, \qquad \Delta_{\mathrm{comp}}(\mathsf{Q}) = \begin{cases} 0, & E^A_t = \varnothing, \\ \max_{e \in E^A_t} \mathrm{UCB}_{\mathrm{sim}}[\delta_e(\mathsf{Q})], & E^A_t \neq \varnothing. \end{cases}$$

Here the simultaneous confidence procedure must cover the family of inspected or adaptively selected edges. A zero defect on every edge defines a compatible assignment for the declared comparison maps. A small defect is only approximate predictive compatibility. It is not causal compatibility unless the node laws are interventional or counterfactual distributions from explicit SCMs and the maps preserve their declared causal semantics.

The closest mathematical prior work for this extension is D'Acunto, Di Lorenzo, and Barbarossa's *Networks of Causal Abstractions: A Sheaf-theoretic Framework* [13]. Their causal abstraction network coordinates heterogeneous causal models using network sheaves and cosheaves, restriction maps, a connection Laplacian, global sections, and mixture causal models. EventFrame adapts the local-to-global compatibility pattern to event-centered predictive laws, then combines it with within-bucket Anti-Pigeon tests, residual-cache certification, and priority-aware staged execution. It does not inherit their causal semantics, consistency results, convergence results, or mixture-learning guarantees.

Accordingly, the EventFrame construction is described only as a sheaf-inspired compatibility scaffold. It should be called a sheaf only after its assigned spaces and restriction maps satisfy the required identity and composition laws. EventFrame does not assume those laws merely because local forecasts are connected by a graph.

EventFrame calls a validated local revision of this scaffold a **predictive sheaf snap**. This is paper-specific terminology, not a standard sheaf-theoretic operation. Write the published compatibility structure at version v as:

$$\Xi_A^{(v)} = \left(\mathcal{G}^{A,(v)}, \{\mathcal{Y}_i\}_{i \in V^{A,(v)}}, \{\mathcal{Y}_e, D_e, w_e, \{g_{ie} : i \in e\}\}_{e \in E^{A,(v)}}\right).$$

For an affected neighborhood $\mathcal{N}$, let D_t^{design} and D_t^{conf} be disjoint chronological design and confirmation blocks satisfying the paper's embargo and as-of rules. Let $\mathfrak{S}_t(\Xi_A^{(v)}; \mathcal{N})$ be a finite, predeclared family of candidate structures constructed only from information available by the slow-path review time, with the unchanged structure $\Xi_A^{(v)}$ included as the no-snap candidate. An edit may split, merge, or duplicate predictive nodes; add or remove predictive-compatibility edges; or select a comparison map from a prevalidated candidate class. The family has bounded neighborhood radius, candidate count, and map complexity. It may not relabel a predictive edge as causal. The notation $\Theta_\Gamma[\Xi']$ means the complete candidate design induced by Ξ', including any required local revision π' of the operational abstraction map, refitted node laws, keys, and certificates. Those dependent components are fitted only on D_t^{design}; a graph edit is never scored while retaining keys or forecasts that are inconsistent with it.

Fix a task-defined comparison-obligation set $\mathfrak{O}_t$ before candidate inspection. Each obligation names local predictions that must remain comparable. A candidate discharges an obligation through a valid direct edge or a composition-valid comparison path; otherwise it must retain the obligation explicitly as unresolved. Let $U_{\text{obl}}(\Xi'; \mathfrak{O}_t) \geq 0$ be the predeclared weighted unresolved burden. This prevents a candidate from obtaining zero defect merely by deleting difficult edges; in particular, an empty graph is not automatically a successful snap when $\mathfrak{O}_t \neq \varnothing$.

Let $\mathcal{D}_{\Delta,t}(\Xi')$ be the reverse dependency closure of the proposed edit, including every bucket, node law, comparison map, key, certificate, cache entry, or edge whose value or validity can change, not only objects edited syntactically. Let $\mathfrak{K}_{\Delta,t}(\Xi')$ and $E_{\Delta,t}^{\text{keep}}(\Xi')$ be its affected active-bucket and retained-or-new-edge projections.

On a chronological design block D_t^{design}, score a candidate by:

$$\begin{aligned}\Psi_t(\Xi'; \Xi_A^{(v)}) = {} & \widehat{\mathcal{R}}_{\text{pri}}^{D_t^{\text{design}}}(\Theta_\Gamma[\Xi']) + \lambda_{\text{comp}} \widehat{\Delta}_{\text{comp}}^{D_t^{\text{design}}}(\Theta_\Gamma[\Xi']; \Xi') \\ & + \lambda_{\text{edit}} d_{\text{edit}}(\Xi', \Xi_A^{(v)}) + \lambda_{\text{snap}} \mathcal{C}_{\text{snap}}(\Xi') + \lambda_{\text{obl}} U_{\text{obl}}(\Xi'; \mathfrak{O}_t),\end{aligned}$$

where the hatted quantities are design-block estimates computed under the predeclared candidate-selection procedure. The compatibility estimate uses the candidate-induced laws and the maps in Ξ'. Every term is finite and normalized to a declared common utility scale, or its coefficient carries the conversion needed to produce that scale. All coefficients are non-negative: λ_{comp} weights compatibility defect, λ_{edit} weights structural churn, and λ_{snap} weights measured or hardware-indexed revision cost. The unresolved-obligation coefficient λ_{obl} is strictly positive unless the candidate family itself requires $U_{\text{obl}} = 0$. Because the candidate family is finite and non-empty, a deterministic tie-breaking rule selects:

$$\Xi_A^{\text{cand}} \in \arg \min_{\Xi' \in \mathfrak{S}_t(\Xi_A^{(v)}; \mathcal{N})} \Psi_t(\Xi'; \Xi_A^{(v)}).$$

Candidate generation and selection do not authorize publication. On D_t^{conf}, define the proper-risk change:

$$\Delta\mathcal{R}^{\mathrm{snap}}_{\mathrm{prop},t} = \widehat{\mathcal{R}}^{D_t^{\mathrm{conf}}}_{\mathrm{prop}}(\Theta_\Gamma[\Xi_A^{\mathrm{cand}}]) - \widehat{\mathcal{R}}^{D_t^{\mathrm{conf}}}_{\mathrm{prop}}(\Theta_\Gamma[\Xi_A^{(v)}]).$$

Let $G^{\mathrm{pri}}_{v\to\mathrm{cand},t}$ be the paired priority-weighted gain defined as in Section 9, and let $C_{v\to\mathrm{cand},t}(h)$ be its resource cost on the same declared utility scale. Declare $\delta_{\mathrm{snap}} > 0$, $\epsilon_{\mathrm{obl}} \geq 0$, and $0 \leq \epsilon^{\mathrm{comp}}_{\mathrm{acc}} \leq \epsilon^{\mathrm{comp}}_{\mathrm{split}}$ before candidate inspection. With $\max\varnothing = 0$, the joint snap-acceptance indicator is:

$$\begin{aligned} A_t^{\mathrm{snap}} = 1 \quad \iff \quad & \mathrm{LCB}\Big[G^{\mathrm{pri}}_{v\to\mathrm{cand},t}\Big] - \mathrm{UCB}[C_{v\to\mathrm{cand},t}(h)] > \delta_{\mathrm{snap}}, \\ & \mathrm{UCB}\big[\Delta\mathcal{R}^{\mathrm{snap}}_{\mathrm{prop},t}\big] \leq \epsilon_{\mathrm{prop}}, \\ & \max_{K\in\mathfrak{K}_{\Delta,t}(\Xi_A^{\mathrm{cand}})} D_K^{\mathrm{cert},\star} \leq \epsilon_{AP}, \\ & \max_{e\in E^{\mathrm{keep}}_{\Delta,t}(\Xi_A^{\mathrm{cand}})} \mathrm{UCB}_{\mathrm{sim}}[\delta_e] \leq \epsilon^{\mathrm{comp}}_{\mathrm{acc}}, \\ & U_{\mathrm{obl}}(\Xi_A^{\mathrm{cand}}; \mathfrak{O}_t) \leq \epsilon_{\mathrm{obl}}. \end{aligned}$$

Set $A_t^{\mathrm{snap}} = 0$ otherwise. Every outcome-dependent quantity in the indicator is computed exclusively from as-of predictions on D_t^{conf}; no confirmation outcome may refit the candidate. The predeclared joint confidence procedure covers every displayed stochastic gate after candidate selection. Repeated reviews use fresh blocks or a sequentially valid procedure. On acceptance, relabel the confirmed candidate as version $v+1$ and write $\mathcal{D}_{\Delta,t} = \mathcal{D}_{\Delta,t}(\Xi_A^{\mathrm{cand}})$. Let $\mathbf{v}^{(v)}$ be the local epoch map, let $\mathcal{C}^{(v)}_{\mathrm{mem}} = (\mathcal{C}^{(v)}_A, \mathcal{C}^{(v)}_R, \mathcal{C}^{(v)}_E)$, let $\mathsf{B}_{\mathcal{D}}$ monotonically increment epochs in dependency closure $\mathcal{D}$, and let $\mathsf{I}_{\mathcal{D}}$ mark every dependent memory entry stale. The publish-or-retain transition is:

$$(\Xi_A^{\mathrm{new}}, \pi^{\mathrm{new}}, \mathbf{v}^{\mathrm{new}}, \mathcal{C}^{\mathrm{new}}_{\mathrm{mem}}) = \begin{cases} \Big(\Xi_A^{(v+1)}, \pi', \mathsf{B}_{\mathcal{D}_{\Delta,t}}(\mathbf{v}^{(v)}), \mathsf{I}_{\mathcal{D}_{\Delta,t}}(\mathcal{C}^{(v)}_{\mathrm{mem}})\Big), & A_t^{\mathrm{snap}} = 1, \\ \Big(\Xi_A^{(v)}, \pi, \mathbf{v}^{(v)}, \mathcal{C}^{(v)}_{\mathrm{mem}}\Big), & A_t^{\mathrm{snap}} = 0. \end{cases}$$

The accepted tuple publishes atomically, and the previous complete version remains available for rollback. The invalidation operator preserves unaffected entries and marks affected ones unusable until recertified; it does not silently assign them certificates under the new version. When $A_t^{\mathrm{snap}} = 0$, no candidate component is published. This operation revises predictive organization only. It may nominate a causal hypothesis, but changing an SCM edge additionally requires the paper's intervention and identification conditions; compatibility improvement alone is insufficient.

When an explicit SCM exists, an accepted predictive snap may nominate a separate finite family of local SCM edits, such as adding or removing an edge, reversing a direction not fixed by temporal order, or introducing a measured mediator or regime variable. Every causal candidate must specify the resulting structural equations, obey declared temporal and domain constraints, and identify the intervention law under which it will be tested. Candidate generation may use predictive defect, but causal selection and confirmation require independent randomized or otherwise identified intervention evidence with correction for the inspected edit family. Observational fit, lower compatibility defect, or successful predictive gluing cannot by themselves orient or promote a causal edge. Until those tests pass, the published causal graph remains unchanged.

The network defect complements rather than replaces Anti-Pigeon. $D_K^\star$ tests hidden external future disagreement inside a bucket; δ_e tests disagreement between representations after both are mapped into a common comparison space. A proposed merge is accepted only when both its external bucket future-diameter and affected edge-defect upper bounds are below their merge thresholds. A bucket or edge is split, invalidated, or routed to deeper review when a lower confidence bound exceeds its split threshold. Separate thresholds $\epsilon_{\text{merge}}^{\text{comp}} < \epsilon_{\text{split}}^{\text{comp}}$ provide hysteresis.

When simple rejection would discard useful local information, a local reconciliation stage may solve:

$$(\overline{\mathsf{Q}}_i)_{i\in\mathcal{N}} \in \arg\min_{(\widetilde{\mathsf{Q}}_i)_{i\in\mathcal{N}}\in\mathfrak{Q}_\mathcal{N}} \left[\sum_{i\in\mathcal{N}} a_i D_i(\widetilde{\mathsf{Q}}_i, \mathsf{Q}_i) + \lambda_A \sum_{e\in E^+(\mathcal{N})} w_e \delta_e(\widetilde{\mathsf{Q}})\right],$$

where $\mathcal{N} \subseteq V_t^A$ is an affected neighborhood, $\mathfrak{Q}_\mathcal{N} \subseteq \prod_{i\in\mathcal{N}} \mathcal{P}(\mathcal{Y}_i)$ is the declared feasible forecast-tuple family, and $E^+(\mathcal{N}) = \{e \in E_t^A : e \cap \mathcal{N} \neq \varnothing\}$ includes both internal and boundary edges. Forecasts outside $\mathcal{N}$ remain fixed and $a_i, w_e \geq 0$. Each $D_i : \mathcal{P}(\mathcal{Y}_i) \times \mathcal{P}(\mathcal{Y}_i) \to [0, +\infty]$ is a declared fidelity divergence with $D_i(Q, Q) = 0$; every divergence and tie-breaking rule is frozen by the evaluation contract. The first term preserves each local forecast; the second penalizes incompatibility without hiding damage at the neighborhood boundary. A minimizer is asserted only when $\mathfrak{Q}_\mathcal{N}$ is compact and the objective is lower semicontinuous, or under another stated existence theorem; otherwise the algorithm must return a declared approximate tuple with an optimality gap. Reconciliation is not unqualified averaging, and the unreconciled forecasts and defects remain available for audit.

For a fixed graph with finite-dimensional embeddings $x_i = \phi_i(\mathsf{Q}_i)$ and linear restrictions R_{ie}, define the boundary operator on edge $e = \{i, j\}$ by:

$$(\partial_A x)_e = R_{ie} x_i - R_{je} x_j, \qquad L_A = \partial_A^* \partial_A.$$

Then $\|\partial_A x\|^2 = \langle x, L_A x\rangle$ and $\ker L_A = \ker \partial_A$, the linearly compatible assignments. If $\lambda_{\max}(L_A) > 0$, the fixed-step refinement

$$x^{(n+1)} = x^{(n)} - \eta L_A x^{(n)}, \qquad 0 < \eta < \frac{2}{\lambda_{\max}(L_A)},$$

converges in finite dimensions to the orthogonal projection of $x^{(0)}$ onto $\ker L_A$. If $L_A = 0$, the assignment is already linearly compatible and no update is required. These statements require a fixed graph, fixed linear restrictions, and the stated inner products. Nonlinear distribution-valued forecasts do not inherit this spectral guarantee automatically.

Finally, a node may represent a predictive regime mixture:

$$\mathsf{Q}_i(\cdot \mid C_t) = \sum_{s=1}^{S_i} \lambda_{is}(C_t) \mathsf{Q}_{is}(\cdot \mid C_t), \qquad \lambda_{is} \geq 0, \quad \sum_{s=1}^{S_i} \lambda_{is} = 1.$$

This mixture can preserve multiple currently plausible mechanisms instead of collapsing them into one habitual prediction. It remains a predictive mixture unless each component has an explicit

SCM and the data and assumptions identify causal interpretation. Mixture learning is the final, most expensive refinement stage; it may revise node laws or comparison maps and then rerun compatibility and reconciliation.

8. Complexity and Runtime Model

EventFrame separates prediction-time computation from post-observation refinement. The fast path may use only $\mathscr{F}_t^{\text{pred}}$ and state S_{t^-}; realized loss, residual estimation, and abstraction learning begin only after the next outcome's availability time.

The reference fast path is:

1. Incrementally update $C_t = e_{t-k+1:t}$.
2. Optionally form $X_t = \chi(C_t, \mathcal{M}_t, G_t, \sigma_t)$.
3. Construct the bounded vector, sheaf-inspired, and as-of graph candidate frontier $\mathcal{N}_t^B$.
4. Check activation and materialized Anti-Pigeon sharing certificates; retrieve the corresponding cached prior and apply only a bounded Bayesian update.
5. Compute baseline law $\mathsf{Q}_B(\cdot \mid C_t)$ and conditional event template $b_t = B(C_t)$, optionally conditioned on the accepted updated belief, or compute packet baseline $B_Y(X_t)$.
6. Construct the bounded action key $k_t = \alpha(C_t)$.
7. Try $\mathcal{C}_{A,t^-}(k_t)$, then $\mathcal{C}_{R,t^-}$, then episodic support if confidence is insufficient.
8. Compose a candidate event output bundle or packet using the separately typed clipped point and law residual components.
9. Evaluate $\mathcal{R}_{\text{pre}}$, confidence, effective support, age, epoch, margin, provenance, and decoder validity from S_{t^-}.
10. Return the admissible prediction or fall back to the baseline. Do not evaluate realized prediction loss yet.

The packet names memory nodes, graph edges, retrieval lane, compaction risk, response mode, and an optional control branch. It predicts what the runtime should read or do; the event prediction describes what is expected to happen.

Expected constant-time lookup is a conditional implementation property. Let T_K be key-construction cost, T_A exact-key lookup, $T_R(N)$ general residual retrieval, $T_E(M)$ episodic retrieval, $T_\oplus$ typed composition, and T_B^{fast} the selective Bayesian work. Then:

$$T_{\text{fast}} = T_C + T_B^{\text{fast}} + T_B(k) + T_K + T_A + I_R T_R(N) + I_E T_E(M) + T_\oplus + T_{\text{pre}},$$

where $I_R, I_E \in \{0,1\}$ indicate fallbacks. Let $B_t^A = |\{e \in \mathcal{N}_t^B : J_t^{\text{act}}(e) = 1\}|$, let M_B bound the updated sufficient-statistic or discrete-hypothesis dimension, and let R_B bound retained changepoint run-length states. For a conjugate, finite-hypothesis, or otherwise bounded primitive,

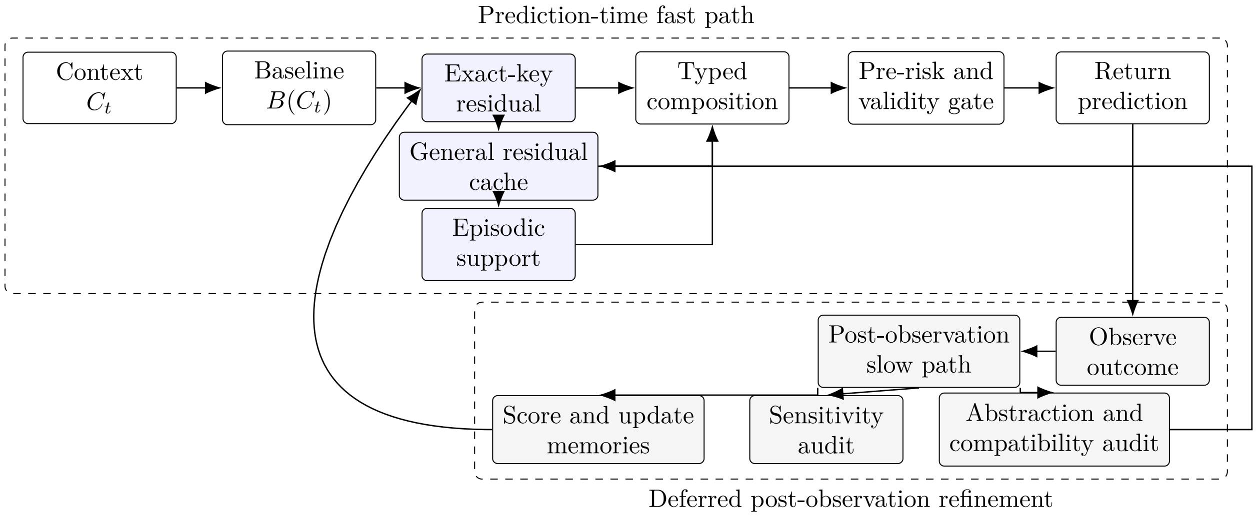

Figure 1: Reference runtime lifecycle. Realized loss and learning occur only after the outcome is observed.

$$T_B^{\text{fast}} = T_{\text{vec}}(k_v) + T_{\text{expand}}(d_{\text{sh}}, d_G) + O(B_t^A M_B R_B) + T_{\text{cert}}.$$

This is history-independent only when k_v, sheaf-inspired degree d_{sh}, as-of graph degree d_G, B_t^A, M_B, and R_B are bounded and when the vector-index query itself has a declared bound. Constant time per sample in a cited streaming algorithm means constant with respect to accumulated stream length under that algorithm's fixed resources; it does not mean zero dependence on particle count, parameter dimension, optimization iterations, graph degree, or hardware. Sliding-window maintenance gives $T_C = O(1)$. A bounded, already-constructed key and bounded hash table give expected $T_A = O(1)$. The claim fails if key construction scans unbounded context, graph degree grows, the posterior or run-length support expands without cap, the table is unbounded, or lookup falls back to unrestricted nearest-neighbor search. Concurrency, hashing, collision handling, selection-probability evaluation, and eviction costs must be measured rather than hidden inside the constant.

The slow path starts after Z_{t+1} or the audited packet target exists:

1. Evaluate $\mathcal{L}_{\text{pred}}$, $\mathcal{L}^H_{\text{event}}$, and $\mathcal{A}_{\text{post}}$.
2. Evaluate packet component loss when a packet was used.
3. Estimate observed residuals and update support/confidence.
4. Consolidate episodic and residual memory.
5. Evaluate inactive audit samples, omitted influence, posterior calibration, and changepoint triggers.
6. Refit or expand Bayesian posteriors with particle, variational, or model-comparison methods when required.
7. Run validity-constrained sensitivity tests.

8. Run causal analysis only when an explicit SCM and identification strategy exist.
9. Audit bucket coverage and future-diameter estimates.
10. Accept split, merge, posterior-sharing, or ontology changes only on independent held-out evidence.

A cost decomposition is:

$$T_{\text{base}} = T_{\text{score}} + T_{\text{residual}} + T_{\text{consolidate}} + T_{B,\text{audit}} + T_{\text{cp}} + \sum_{q=1}^{M_f} T^{(q)}_{\text{predict}} + T_{\text{audit}}$$

$$T_{\text{upgrade}} = T_{\text{comp}} + T_{\text{reconcile}} + T_{\text{snap}} + T_{\text{spectral}} + T_{\text{mixture}} + T_{B,\text{deep}},$$

$$T_{\text{slow}} = T_{\text{base}} + T_{\text{upgrade}} + \sum_{q=1}^{M_c} T^{(q)}_{\text{causal}},$$

where $M_f, M_c \in \mathbb{N}_0$ are the numbers of fuzzing-prediction and causal-analysis invocations. Set $M_c = 0$ when no causal model is available. Slow work must be budgeted, deferred, or batched so it does not silently migrate into the latency-critical path.

The Bayesian upgrade has an orthogonal cumulative ladder that does not renumber the abstraction-refinement stages:

$$\mathcal{B}_0 = \text{bounded activation, certificate lookup, and cached update,}$$
$$\mathcal{B}_1 = \text{bounded robust changepoint monitoring,}$$

$$\mathcal{B}_2 = \text{declared event-pattern forecast refinement,}$$
$$\mathcal{B}_3 = \text{particle, variational SMC, and model recalibration.}$$

Only $\mathcal{B}_0$, and $\mathcal{B}_1$ when its run-length state is explicitly bounded, may be admitted to the direct fast path. $\mathcal{B}_2$ is fast only for a bounded precompiled pattern family and state space. $\mathcal{B}_3$ is slow-path work. A changepoint, high omitted-influence certificate, missing prior, invalid sharing certificate, high-priority case, or exhausted approximation budget escalates to a predeclared deeper Bayesian stage.

The full upgrade is defined as a staged family rather than one indivisible algorithm. Let S_t contain the current forecasts, caches, abstraction graph, audit state, and hardware profile h. Define refinement operators:

$$\mathcal{U}_0 = \text{certified baseline/residual reuse,} \qquad \mathcal{U}_1 = \text{edge compatibility audit,}$$

$$\mathcal{U}_2 = \text{local reconciliation,} \qquad \mathcal{U}_3 = \text{bounded predictive sheaf snap,}$$

$$\mathcal{U}_4 = \text{component or spectral refinement}, \qquad \mathcal{U}_5 = \text{regime-mixture and map refinement}.$$

Starting from $S_t^{(0)} = \mathcal{U}_0(S_{t^-})$, let $r_n \in \{1, 2, 3, 4, 5\}$ be the stage selected for slow-path invocation n, subject to its prerequisites. The step-integration recurrence is:

$$S_t^{(n)} = \mathcal{U}_{r_n}(S_t^{(n-1)}), \qquad n = 1, \ldots, N_t.$$

Let every conservative invocation-cost bound be strictly positive, $c_r^U(h, S) > 0$, and charge repeated stages separately:

$$C_t^U(n; h) = \sum_{q=1}^{n} c_{r_q}^U(h, S_t^{(q-1)}).$$

Invocation n is permitted only if:

$$C_t^U(n; h) \le \mathcal{B}(p_t^{\text{pri}}),$$

and all prerequisite evidence and safety gates pass. The run stops at the first failed budget or prerequisite check, a declared convergence condition, or a finite iteration cap. Its reported refinement depth is:

$$d_t(h) = \max\left(\{0\} \cup \{r_1, \ldots, r_{N_t}\}\right).$$

Here $p_t^{\text{pri}} \in [0, 1]$ is priority declared from prediction-time information, $\mathcal{B}$ is a priority-dependent resource budget, and $c_r^U(h, S)$ is a preregistered upper confidence bound or deterministic worst-case bound on hardware profile h. The runtime also accumulates actual cost and reports overruns. Predicted admission alone is not a hard budget guarantee; a strict deadline additionally requires interruptible stages and a reserved worst-case completion margin or a deterministic stop. Stage 5 may revise mixtures or comparison maps, after which Stages 1–4 may be selected again; every rerun appears again in C_t^U. The architecture targets certified reuse plus all five refinement stages; $d_t(h)$ records the deepest stage reached, while the complete invocation sequence $(r_1, \ldots, r_{N_t})$, actual cost, and stopping reason are also reported.

This definition separates semantic interfaces from hardware policy. Faster processors, larger memory, improved accelerators, or cheaper distributional solvers reduce measured costs and their conservative bounds and can increase $d_t(h)$ without changing event, residual, compatibility, or causality definitions. A conforming implementation must therefore record both the output stage and the hardware/cost profile used to select it.

For a changed edge set E_Δ, compatibility work is approximately:

$$T_{\text{comp}} = O\left(\sum_{e \in E_\Delta} C_{D_e}\right),$$

where C_{D_e} is the cost of mapping and comparing the two incident forecasts. With bounded local degree this is local in the changed neighborhood. Spectral work depends on component size, representation dimension, sparsity, solver, and requested tolerance. Mixture refinement additionally depends on component counts and optimization restarts and is expected to remain the most expensive stage. No fixed millisecond or slowdown claim is made without an implementation and hardware profile.

For a finite predictive-snap family $\mathfrak{S}_t$, the design-block computation is bounded by the work charged for every inspected candidate:

$$T_{\text{snap}} \le T_{\text{generate}} + \sum_{\Xi' \in \mathfrak{S}_t} \left[T_{\text{refit}}(\Xi') + \sum_{e \in E_\Delta(\Xi')} C_{D_e} + T_{\text{obl}}(\Xi') + T_{\text{score}}(\Xi') \right] + T_{\text{confirm}} + T_{\text{publish}}.$$

Here T_{generate} includes bounded neighborhood and candidate construction, T_{obl} validates direct or composed comparison obligations, and T_{confirm} is confirmation scoring cost rather than the wall-clock wait for future outcomes. T_{score} includes candidate risk, affected-bucket Anti-Pigeon evaluation, and affected-edge compatibility evaluation not already charged in the explicit edge sum; an implementation must partition these measurements so no operation is omitted or counted twice. The ordinary T_{comp} term audits the published graph, whereas the inner edge sum charges incremental candidate comparisons. The untouched confirmation block may delay publication but is not placed on the current prediction path. Candidate count, neighborhood radius, reverse dependency closure, refit budget, comparison-obligation set, and map class must be bounded before the review begins; unrestricted graph-structure search is not a conforming snapping implementation. The candidate graph, induced local abstraction mapping, and dependent keys are built in shadow state, and publication is an atomic graph-key-epoch swap. Consequently snapping requires only the existing version-consistent epoch check on the direct fast path. The indirect cost is a temporary rise in baseline or certified-fallback use while affected residual entries are recertified.

The integration roadmap is cumulative:

1. Specify typed Bayesian evidence, parameter spaces, activation maps, source model, selection semantics, bounded sufficient statistics, and deterministic fallbacks.
2. Add shadow-only activation, independent audit sampling, and omitted-influence measurement before allowing production posterior updates.
3. Materialize Anti-Pigeon posterior-sharing certificates and enable bounded cached updates with atomic posterior-key-epoch publication.
4. Add bounded robust changepoint monitoring and targeted invalidation; keep particle or unbounded run-length methods asynchronous.
5. Add read-only compatibility auditing and materialize epoch/margin certificates for the unchanged residual fast path.
6. Enable local reconciliation only on held-out evidence that it improves priority-weighted utility without unacceptable harm.
7. Add bounded predictive sheaf snapping with shadow construction, targeted invalidation, atomic publication, and rollback.

8. Add component-level spectral diagnostics and refinement where linearization assumptions are validated.
9. Add predictive regime mixtures and deep Bayesian state-space refinement; promote causal interpretations only with explicit SCMs and identification assumptions.
10. Rebenchmark every stage on each hardware generation and widen activation budgets without weakening validation, selection, or Anti-Pigeon gates.

The runtime reports prediction score, event-aware timing error, pre-risk calibration, cache hit and fallback rates, residual improvement, activation and audit rates, selected and unselected posterior calibration, omitted influence, effective support, changepoint delay and false alarms, Bayesian frontier size, posterior-update cost, decoder failures, slow-path delay, selected Bayesian and abstraction refinement depths, hardware profile, edge defects, bucket audit coverage, snap attempts and acceptances, rollback, cache recertification delay, and split/merge churn. Without these measurements, the claimed fast/slow tradeoff remains an architectural proposal rather than an established result.

9. Experimental Evaluation

EventFrame's main claims require experiments. The framework should be evaluated on whether compressed event frames preserve intervention-relevant distinctions, whether structured events improve interpretability and prediction, whether residual caches reduce cost or error, whether property fuzzing discovers stable invariants, and whether abstraction preserves target-relevant transition behavior.

Every experiment follows one leakage-resistant protocol. Raw trajectories and the target definition are fixed before candidate resolutions or abstractions are compared. Evaluation uses rolling-origin or forward-chaining windows, grouped by independent trajectory or entity. Between training/design and evaluation blocks, impose an embargo at least as long as the maximum context span plus forecast horizon plus outcome-label delay. At every prediction, replay only S_{t^-} and objects with availability time at most t; delayed corrections, cache entries, confidence updates, epochs, audit results, and outcomes are unavailable until their recorded availability times.

All learned preprocessing, feature normalization, priority models, temporal-resolution choices, thresholds, perturbation generators, and policy-selection rules are fitted inside the corresponding training or design window. Candidate selection and all iterative analysis use $\mathcal{S}_{\text{obj}}$, drawn under P_{obj}. After those choices and the analysis code are frozen, confirm final claims once on untouched $\mathcal{S}_{\text{conf}}$, drawn under P_{conf}. Confidence intervals and tests use trajectory clusters, blocked resampling, or a justified effective sample size rather than treating overlapping contexts as independent. Repeated monitoring uses confidence sequences, alpha spending, or preregistered review times. Random example-level splits are not admissible when contexts overlap in time, entity, source episode, or label construction.

A minimal synthetic event world should generate trajectories with known transition rules. Each event should expose the fields:

$$e_t = (w_t, a_t, \tau_t, \ell_t, m_t, h_t, x_t, c_t).$$

The generator should include many microscopic variables but control which variables actually influence event timing or downstream state. It should also allow multiple temporal resolutions, such as seconds, milliseconds, and microseconds. This makes it possible to test whether coarse-graining preserves intervention-effective distinctions, whether fuzzing recovers true dependencies, and whether abstraction removes irrelevant detail without damaging prediction.

The first experiment measures marked next-event prediction. Compare:

1. A baseline predictor without residual cache.
2. A baseline predictor with episodic retrieval.
3. A baseline predictor with residual cache.
4. A full EventFrame reference predictor.

The primary metric is the untransformed proper score $\mathcal{L}_{\text{pred}}$, including the no-event outcome and event identity, with confidence intervals over trajectories. Report $\mathcal{L}^{H}_{\text{event}}$, mark accuracy, calibration, censoring performance, and auxiliary-field loss as separately frozen diagnostics. A residual method may claim proper-score improvement only for a law component with declared kernel $\mathfrak{K}^{Q}_{H}$; a point-only record is evaluated only on point diagnostics, and a joint record must be evaluated on the complete bundle. The key question is whether distributional residual composition improves held-out forecasts without purchasing that gain through harmful template fields.

The second experiment tests compression and target relevance. Define a finite candidate distinction set $\mathcal{D}_t$, fit every full and ablated design before confirmation, vary Γ_{Δ_τ}, and report the simultaneous-confidence-classified ablation ratio $s^{\text{pred}}_{\text{eff}}$. Confirmation outcomes may classify frozen distinctions but may not regenerate or refit them. In a synthetic generator, randomized changes to known structural variables can additionally identify and report $s^{\text{causal}}_{\text{eff}}$. Observational benchmarks report only the predictive ratio unless a separate identification argument is supplied.

The third experiment measures cache utility under as-of replay. Report action-residual hit rate, general residual hit rate, post-hit temporal loss, baseline temporal loss on the same examples, confidence calibration, effective support, cache age, epoch and margin rejection, provenance rejection, and the fraction of hits that improve prediction. A residual cache is useful only if retrieved residuals improve over the baseline often enough to justify lookup and maintenance. Cache pollution should be measured by tracking entries that repeatedly fail to improve predictions. For the action-residual path, also report how often expected $O(1)$ lookup succeeds without falling back to nearest-neighbor residual search or episodic retrieval.

The fourth experiment evaluates selective Bayesian updating and Anti-Pigeon posterior granularity. Compare no Bayesian update, update-all Bayesian inference, selective activation with naive unconditioned likelihood, selective inference with the declared selection-conditioned likelihood, selective inference with separate per-event posteriors, Anti-Pigeon-certified posterior sharing, and a simulation-only oracle frontier. The generator should include informative activation, correlated sources, hidden divergent subgroups, abrupt and gradual regime changes, and relevant events outside the normal frontier.

Primary forecast evaluation remains on the complete chronological event stream, not only activated cases. Report untransformed proper score, calibration and interval coverage on the full stream and activated stratum, activation precision and recall for outcome-relevant evidence, missed high-priority events, false posterior merges, posterior fragmentation, effective sample size, changepoint false alarms and delay, posterior invalidation delay, and U_t^{omit} coverage. Conditional on inactivity, the audit draw must be independent of activation-score magnitude under its frozen schedule; compare its expanded forecast with the local forecast and report audit rate, final inclusion probabilities, design-weight stability, and reservoir saturation. Results from naive selective updating must be labeled as activation-conditioned working-posterior results unless conditional ignorability is established.

Runtime reporting includes frontier size, activated count, posterior-hypothesis count, run-length support after pruning, cache hit rate, memory, and 50th, 95th, and 99th percentile latency for $\mathcal{B}_0$ through $\mathcal{B}_3$. Evaluate exact or near-exact streaming updates against capped approximations on short sequences where an oracle is computable. A fixed-resource claim passes only if vector retrieval, graph degrees, active candidates, hypotheses, update rank, audit reservoir, and changepoint state are all bounded and approximation error remains within its preregistered tolerance.

The fifth experiment evaluates property fuzzing. For each field ϕ_i, perturb it across a declared range and compute:

$$S_g = \min\left(1, \frac{\Delta_g}{\eta_g}\right).$$

The experiment should compare discovered stable fields to the known generating rules. If the generator makes location irrelevant to timing, temporal fuzzing should identify location as stable for that target. If the generator makes actor identity relevant, actor perturbation should change temporal predictions beyond threshold.

The sixth experiment tests ontology review. Deliberately misassign generated fields and use $I_{i\to g}^{\mathrm{model}}$ to nominate retain, migrate, split, or uncertain states. Report recovery of predictive roles. Evaluate causal-role recovery only in generators whose structural equations and randomized interventions are known.

The seventh experiment evaluates confluence, divergence, and audit coverage. Each group retains a traceability frame and a coverage-aware context audit set. Place hidden divergent contexts outside the medoid neighborhood and measure false-negative rate as audit-set size and context-space coverage change. A one-representative baseline should be included to demonstrate why one anchor is insufficient.

The eighth experiment evaluates invariant stability over time. Candidate invariants discovered in one trajectory segment should be tested on later segments and under distribution shift. This distinguishes local accidental stability from robust invariance. Report the rate at which candidate invariants remain valid, fail, or become conditional, following the distinction between abrupt, gradual, recurring, and other drift patterns surveyed in [12].

The ninth experiment estimates $\varepsilon^{\star}_{\mathrm{lump}}(\pi)$ over pairs of contexts sharing an operational key and reports a simultaneous upper confidence bound. For each bucket K, compare $\widehat{D}^{\star}_K$ with the known external $D^{\star}_K$ in synthetic data, and report the model-only diagnostic D_K^{mdl} separately. This directly tests false merges, divergence missed by incomplete audit coverage, and the failure mode in which a uniformly wrong model falsely appears internally consistent.

The same experiment includes an observed regime shift $\zeta_a \to \zeta_b$. Measure $D^{\mathrm{reg}}_{i,a,b}$, post-loss increase, detection delay, false alarms, and adaptation cost. A separate randomized generator test may establish whether the regime variable is causal; ordinary conditional divergence may not.

The tenth experiment evaluates runtime tradeoffs. Measure fast-path latency, slow-path cost, cache update cost, Bayesian update cost, and memory growth. Report the conditions under which residual lookup and bounded Bayesian updating approximate constant-time behavior and the conditions under which either fails.

The eleventh experiment evaluates predictive sheaf snapping. Synthetic event networks should include known local miswirings, unnecessary edges, missing regime splits, and deliberately misleading comparison maps. Compare no snapping, the bounded preregistered snap family, a larger-family stress test, and an oracle edit available only in simulation. Candidate generation and selection use the chronological design block; acceptance uses a later untouched block. Report beneficial-, harmful-, false-, and missed-snap rates; edit distance; compatibility-defect change; external future-diameter; unresolved comparison-obligation burden; ordinary and priority-weighted proper-score change; graph churn; rollback frequency; candidate count; selection and confirmation cost; affected-cache and posterior fraction; cache-hit recovery time; and fast-path latency before, during, and after publication. Report whether simultaneous coverage remains calibrated under adaptive edge and candidate inspection. Causal-edge recovery is scored only in generators with explicit SCMs, candidate structural equations, and identified interventions; it must compare predictive nomination with causal acceptance so that a useful predictive snap is not counted as a correct causal edit merely because its forecast improved.

The twelfth experiment evaluates complete staged-execution policies, not merely adjacent stage labels. A policy q freezes its admissible invocation sequences, prerequisites, repetition rules, stopping rule, and cost bound before confirmation. Compare preregistered policies including $\mathcal{U}_0$ alone, cumulative one-pass policies through each later stage, the Bayesian ladder $\mathcal{B}_0, \dots, \mathcal{B}_3$, and any adaptive policy that may repeat or reorder stages. Report the realized invocation sequence for every case, ordinary proper score, edge-defect calibration, high-priority false-negative rate, probability of harmful correction, snap acceptance and rollback, posterior invalidation, split/merge churn, budget overruns, and slow-path latency at the 50th, 95th, and 99th percentiles. Results from one policy do not establish the value of another.

Average correction alone is not the deployment criterion. On a non-empty evaluation set, let $p_t^{\text{pri}} \in [0,1]$ be assigned before the outcome by a rule frozen independently of the stages being compared, let $w_{\text{pri}}(p) > 0$ be a declared finite importance function, and normalize over evaluation cases:

$$\widetilde{w}_t = \frac{w_{\text{pri}}(p_t^{\text{pri}})}{\sum_{u=1}^{T} w_{\text{pri}}(p_u^{\text{pri}})}.$$

For complete policies q_a and q_b, define the bounded per-case system losses by:

$$L_t^{[q]} = \mathcal{A}_{\text{post}}(\mathcal{O}_t^{[q]}, Z_{t+1}) \in [0,1].$$

The untransformed proper score is reported separately. Priority-weighted absolute gain is:

$$G_{a \to b}^{\text{pri}} = \sum_{t=1}^{T} \widetilde{w}_t \left(L_t^{[q_a]} - L_t^{[q_b]} \right),$$

When the weighted baseline-loss denominator is strictly positive, priority-weighted relative risk reduction is:

$$G^{\mathrm{pri}}_{a\to b,\mathrm{rel}} = \frac{\sum_{t=1}^{T} w_{\mathrm{pri}}(p_t^{\mathrm{pri}})\left(L_t^{[q_a]} - L_t^{[q_b]}\right)}{\sum_{t=1}^{T} w_{\mathrm{pri}}(p_t^{\mathrm{pri}}) L_t^{[q_a]}}.$$

If $\sum_t w_{\mathrm{pri}}(p_t^{\mathrm{pri}}) L_t^{[q_a]} = 0$, the relative statistic is undefined and the experiment reports only absolute gain and the paired loss distribution.

Priority must not be assigned after seeing whether a stage helped, and a candidate stage may not control the rule that weights its own evaluation. Report unweighted results beside weighted results, the full priority-stratified loss distribution, and uncertainty intervals. A small average gain may justify a stage if it produces a credible reduction in predeclared critical-case failure with bounded harm elsewhere.

Latency percentage and loss percentage are not directly commensurate. For hardware profile h, choose $T_{\mathrm{budget}} > 0$ and non-negative conversion coefficients, then convert measured resource effects into the same declared utility scale:

$$C_{a\to b}(h) = \lambda_T \frac{\Delta T_{a\to b}(h)}{T_{\mathrm{budget}}} + \lambda_C \Delta C^{\mathrm{compute}}_{a\to b}(h) + \lambda_M \Delta C^{\mathrm{memory}}_{a\to b}(h).$$

The two ΔC terms are declared normalized changes, not raw processor operations or bytes. Their coefficients and λ_T convert all three resource terms into the same utility units as the gain statistic.

An evidence-controlled promotion rule may conservatively require a paired lower confidence bound on gain and an upper confidence bound on measured resource cost:

$$\mathrm{LCB}_{\mathrm{paired}}[G^{\mathrm{pri}}_{a\to b}] - \mathrm{UCB}[C_{a\to b}(h)] > \delta_{\mathrm{safety}},$$

Promotion also requires the paired upper confidence bound on proper-score degradation of q_b relative to q_a to be at most the preregistered ϵ_{prop}.

or a joint confidence construction with the same coverage, or a separately preregistered critical-risk constraint. If stages, priorities, thresholds, or hardware profiles are selected after inspecting the same evaluation data, the confidence procedure must adjust for those comparisons or use a fresh confirmation set. The weights, normalization, confidence procedure, safety margin, and hardware profile must be fixed before evaluation. This is a proposed decision rule, not evidence that any upgrade stage currently passes it.

Ablation studies should remove one component at a time: residual cache, episodic memory, Bayesian activation, selection correction, Anti-Pigeon posterior sharing, inactive-event audit, changepoint invalidation, fuzzing, abstraction, compatibility audit, reconciliation, predictive sheaf snapping, targeted invalidation, rollback, spectral refinement, regime mixtures, and slow-path refinement. The paper should treat negative results as informative. If selective updating loses calibration or misses important omitted evidence, the frontier or selection model fails. If residual caches fail in a domain, the failure helps characterize when EventFrame is useful. If fuzzing produces unstable invariants, the thresholds or perturbation families may be wrong. If snapping reduces design-block defect but harms untouched proper score or causes persistent cache-hit collapse, the snap policy fails its stated purpose.

The evaluation plan is deliberately falsifiable. Each claim should be tied to a measurable result. The next section lists open problems that remain even if the initial experiments succeed.

Discussion: Innovation and Scientific Refinement

EventFrame treats refinement conservatively. A residual, anomaly, or fuzzing result first identifies a predictive distinction. It becomes a causal distinction only when randomized or otherwise identified intervention evidence supports that interpretation.

The runtime alternates between compression and refinement. Lumpability asks when detailed distinctions can be removed because future behavior remains equivalent for the target. Anti-Pigeon asks when an abstraction hides incompatible futures and must split. Validity-constrained perturbation supplies model-sensitivity evidence; an explicit causal model is required for causal intervention claims.

The alternation can be written operationally:

1. Predict with the current event ontology and abstraction.
2. After observation, measure $\mathcal{A}_{\text{post}}$.
3. If forward-held-out post-loss remains low and proper-score non-inferiority holds, preserve the current abstraction.
4. If post-loss remains high, run sensitivity and abstraction audits.
5. If distinctions do not affect the target, compress through lumpability.
6. If distinctions repeatedly affect the target, refine through Anti-Pigeon or ontology revision.
7. If incompatibility is localized to a heterogeneous abstraction neighborhood, generate a bounded predictive sheaf-snap family in shadow state.
8. Publish a snap only after untouched chronological confirmation; otherwise retain the current graph.

EventFrame does not assume that its ontology is correct at the start. The ontology is a working compression that earns predictive stability on independent tests. A predictive snap may reorganize nodes, compatibility edges, or comparison maps, but causal-edge credibility is evaluated separately under an SCM or identified intervention design.

This discussion also limits the claim. EventFrame does not provide a theory of scientific discovery. It provides a runtime vocabulary for prediction, residual diagnosis, sensitivity testing, and evidence-controlled abstraction.

Convergence requires stronger conditions than stationarity and finite move types. Consider a finite set $\mathfrak{S}$ of complete candidate abstraction states evaluated on a fixed validation distribution. Let

$$\Phi(s) = \sum_{t=1}^{T} \widetilde{w}_t \mathcal{A}^{s}_{\mathrm{post},t} + \lambda_{\mathrm{rep}} \mathcal{C}_{\mathrm{rep}}(s),$$

where the normalized priority weights $\widetilde{w}_t$ are fixed with the evaluation set and infeasible Anti-Pigeon states are excluded. If the update rule is deterministic and accepts $s \to s'$ only when $\Phi(s') \leq \Phi(s) - \delta$ for a fixed $\delta > 0$, then no state can be revisited and the process terminates after at most $|\mathfrak{S}| - 1$ accepted moves at a state with no improving candidate move. This is a finite-state descent result, not a guarantee for an online changing environment. With noisy estimates, adaptive candidate generation, changing caches, or distribution drift, the result does not apply unless confidence bounds and a fixed potential restore the strict-decrease invariant.

The next section lists open problems that remain before this pattern can support stronger guarantees.

10. Open Problems

EventFrame is a framework, not a completed theory. Several open problems must be resolved before it can support strong claims.

The first open problem is the status of substrate-to-frame compression. EventFrame hypothesizes that predictively effective distinctions have a small held-out ablation ratio in compressible domains. A separate causal sparsity ratio is meaningful only where interventions are identified. Physical information bounds provide a limiting analogy only for physical substrates; they do not justify compression in simulated or software systems. A future theory would need to state when a coarse-graining Γ preserves exactly the distinctions needed for prediction and, where applicable, intervention.

The second open problem is formal guarantees. The paper now specifies a finite-dimensional operator space, clipping, admissible projection, and decoder, but does not prove that learned encoders preserve semantic fields or that non-convex admissible projections are stable. The CFS connection remains structural inspiration, not physical equivalence.

The third open problem is online convergence. Finite-state strict descent terminates on a fixed evaluation distribution, but real runtimes change caches, candidates, and data distributions. Regret, tracking error, and churn bounds under drift remain open.

The fourth open problem is event scoring. Proper marked-event scores handle identity, time, uncertainty, and censoring, but practical systems still need calibrated component distances over actors, locations, mechanisms, and auxiliary state.

The fifth open problem is grounding. EventFrame assumes that event fields can be extracted or inferred. In many domains, this is difficult. The ”why” and ”how” fields may be ambiguous, contested, or unavailable. Confidence metadata can record uncertainty, but it does not solve extraction. A robust system must distinguish observed fields from inferred fields and must avoid treating speculation as fact.

The sixth open problem is drift. Residual caches depend on the assumption that similar contexts continue to produce similar transition errors. When the environment changes, old residuals may become harmful. Cache metadata, decay, and slow-path review can reduce this risk, but drift detection remains a core challenge.

The seventh open problem is cache pollution. If the system stores too many residuals, it may memorize noise. If it stores too few, it misses useful corrections. The right update rule may depend on domain, context length, confidence, and the cost of false correction.

The eighth open problem is residual confidence under drift. Residuals are statistical corrections, not causal hypotheses. A stronger theory would specify decay schedules, effective sample size, false-correction costs, and change detection.

The ninth open problem is robust invariant extraction. Fuzzing can identify candidate invariants, but perturbation validity is hard. A counterfactual event may be syntactically valid but semantically impossible. Thresholds may be too permissive or too strict. Invariants may be local, conditional, or unstable under distribution shift.

The tenth open problem is abstraction quality. Approximate predictive lumpability is attractive, but exact lumpability is usually too strong. The framework needs practical criteria for deciding when an abstraction is good enough for one target but unsafe for another. An abstraction that preserves timing may destroy causal explanation.

The eleventh open problem is confluence and divergence detection. A system needs criteria for deciding when event streams have truly become prediction-equivalent and when small distinctions are about to amplify. Bad confluence loses necessary distinctions; bad divergence preserves noise as if it were signal.

The twelfth open problem is audit-set construction. One traceability frame is necessary but insufficient. Future work should compare coresets, boundary examples, reservoir sampling, coverage metrics, and adversarial audits.

The thirteenth open problem is temporal resolution selection. Finer time precision can create more candidate frames and expose divergence boundaries, but it can also increase noise, cache pressure, and false distinctions. The framework needs principled methods for choosing Δ_τ, possibly adapting it across domains or event groups.

The fourteenth open problem is multimodal scaling. Event frames may be built from text, sensor streams, images, logs, graphs, or simulations. A unified event representation must allow these sources to contribute without pretending that all fields have the same reliability or comparison rule.

The fifteenth open problem is evaluation design. Synthetic worlds are useful because ground truth is known, but real domains are messier. A credible research program should move from synthetic tests to controlled real-world benchmarks while preserving the ability to inspect fields, residuals, and invariants.

The sixteenth open problem is causal identification. Model graph perturbations measure sensitivity. Future work must specify structural equations, intervention targets, identification assumptions, and transport conditions before promoting predictive dependencies to causal edges.

The seventeenth open problem is empirical evidence. The bibliography and mathematical distinctions are now explicit, but implementation, ablation, and controlled real-world validation remain absent.

The eighteenth open problem is compatibility-map validity. Pairwise comparison maps may be learned incorrectly, may fail to compose, or may erase exactly the distinctions that Anti-Pigeon is intended to protect. A graph of forecasts is not automatically a sheaf, and low edge defect under bad maps is not evidence of global coherence.

The nineteenth open problem is predictive sheaf-snap search. Candidate families must be expressive enough to repair local incompatibility but bounded enough to avoid combinatorial search, repeated-test overfitting, and graph churn. Theory is needed for neighborhood selection, edit penalties, rollback, and cache-hit recovery under drift.

The twentieth open problem is priority calibration. Priority weighting can protect rare consequential cases, but a misspecified or manipulable priority function can hide ordinary harms or overfit a favored subgroup. Priority must be assigned before outcomes and evaluated beside unweighted and stratified results.

The twenty-first open problem is hardware-aware scheduling. The staged architecture permits deeper refinement as hardware improves, but stage-cost prediction, queue stability, energy use, worst-case deadlines, and post-snap cache recovery remain implementation-dependent. Faster hardware does not relax statistical, causal, or safety prerequisites.

The twenty-second open problem is selective posterior calibration. Activation depends on relevance, novelty, topology, and source structure, so the admitted stream is generally not an ignorable sample. Selection-conditioned likelihoods, inverse-probability methods, doubly robust audits, and conservative working-posterior semantics should be compared under misspecification.

The twenty-third open problem is posterior granularity. Anti-Pigeon supplies an external divergence gate for sharing, but optimal split and merge policies under sparse evidence, multiple horizons, source dependence, and drift remain unknown. Over-sharing creates confident category errors; over-splitting wastes evidence and memory.

The twenty-fourth open problem is bounded changepoint inference. Exact Bayesian online changepoint support grows with stream length. Truncation, pruning, and finite-state approximations require error bounds that remain meaningful under selective activation and delayed labels.

The twenty-fifth open problem is omitted influence. A bounded local frontier can miss weak individual signals whose joint effect is material. Independent inactive-event audits estimate this risk only on sampled candidates; coverage guarantees under adversarial or highly correlated omissions remain open.

These open problems define the boundary of the current paper. The framework is useful if it makes prediction, memory, and abstraction more explicit and testable. It should not be presented as a final cognitive architecture, universal predictor, or complete mathematical theory. The conclusion summarizes the role EventFrame can play as a conservative event-centric substrate.

11. Conclusion

EventFrame proposes typed, task-relative event frames for prediction without treating them as fundamental entities. A coarse-graining Γ_{Δ_τ} maps detailed histories into event frames at a declared resolution. The predictor returns a distribution over marked event times and a no-event outcome, evaluated by a proper score; bounded event-aware timing error remains diagnostic.

At fixed temporal resolution, a population objective defines an oracle benchmark that minimizes expected priority-weighted post-observation action plus non-negative representation cost under external target-law and proper-score constraints. The operational rule is different: it minimizes empirical action over a finite candidate family whose bucket and proper-score constraints have predeclared certificates. Oracle feasibility and empirical certifiability are not interchangeable. Candidate selection and untouched chronological confirmation are separate. A distinct pre-observation risk gates fast-path use from S_{t^-} because realized loss is unavailable until its recorded availability time.

For an in-horizon concrete event, the point component of a typed residual record gives the type-resolved template composition:

$$e_{t+1}^{\mathrm{tmpl}} = b_t \oplus_E \bar{r}_t^E.$$

The point operator encodes events into a finite-dimensional tagged self-adjoint operator space, norm-clips the point residual, projects into a declared admissible set, and decodes with a named decoder. That component is undefined when the originating horizon expires without an event. A separately tagged law component drives a full-outcome Markov kernel, explicitly governs probability flow into and out of the no-event atom, and supplies the law evaluated by the proper score. A fixed decision rule aligns the final mark and time with that law; joint forward validation determines whether the auxiliary template fields also help. Runtime packets use an independent packet encoder, residual space, admissible set, and operator $\oplus_Y$. The construction takes limited inspiration from CFS self-adjoint operator representations; its clipping and projection are EventFrame definitions, not a CFS action or physical theory.

Episodic memory stores prior cases; residual memory stores prior statistical corrections. Residuals are not causal hypotheses without separate intervention evidence. A selective Bayesian frontier adds bounded vector retrieval, sheaf-inspired neighbors, and as-of graph adjacency. A frozen activation score admits only a bounded subset, while Anti-Pigeon decides which admitted events may share a posterior. Informative selection requires a selection-conditioned likelihood; otherwise the result is explicitly only an activation-conditioned working posterior. Independent inactive-event audits bound omitted influence, and changepoint triggers invalidate posterior and residual epochs before slow recalibration.

The fast path performs bounded lookup, a capped cached posterior update, typed composition, and pre-risk checks. The slow path evaluates realized scores, updates confidence, audits inactive evidence, runs model-sensitivity and changepoint procedures, and tests abstractions. Particle filters, variational sequential Monte Carlo, model comparison, and unrestricted recalibration remain slow-path operations unless a concrete implementation supplies hard resource bounds. Causal-edge updates require an explicit structural causal model and identification strategy; an as-of outgoing edge can nominate a candidate but cannot provide future evidence.

Approximate predictive lumpability compares detailed contexts that map to the same operational abstraction key. Anti-Pigeon rejects buckets whose externally estimated target-law future-diameter exceeds threshold; a candidate model's own forecast agreement is diagnostic and cannot certify itself. Every bucket retains a concrete traceability frame, but divergence testing uses a coverage-aware context audit set because one representative cannot characterize a heterogeneous group. Observed regime divergence is evaluated on common support and supports predictive adaptation, not causal attribution by itself.

The target architecture also admits a staged abstraction compatibility network. It begins with certified residual reuse, then adds edge audits, local reconciliation, bounded predictive sheaf snapping, spectral refinement under declared linear assumptions, and predictive regime mixtures. A snap selects from a finite local edit family, preserves externally fixed comparison obligations, requires later untouched confirmation, and publishes an atomic graph-key-epoch version with targeted cache invalidation and rollback. It reorganizes predictive compatibility and does not establish causality. Hardware improvements may permit a greater refinement depth, but do not change the mathematical interfaces or waive evidence gates. Rare high-priority corrections are evaluated with predeclared priority-weighted risk alongside unweighted and stratified results.

A finite-state abstraction search terminates under a strict-decrease rule on a fixed potential and fixed evaluation distribution. This result does not imply convergence in an online drifting environment. Implementation, staged ablation, compatibility-map validation, audit-coverage studies, and controlled real-world validation remain necessary before the framework's utility claims can be accepted.

Appendix A. Symbol Index

This index resolves the core symbols used by the formulas. Component spaces for event fields use calligraphic letters without descriptive subscripts; packet component spaces always carry descriptive subscripts.

Ω: substrate state space. It is never used as a cost function.

A_t, ω_{A_t}: finite substrate/computational region and its history.

Δ_τ, Γ_{Δ_τ}: temporal resolution and task-relative coarse-graining map.

e_t, C_t, $\mathfrak{C}_{\mathrm{adm}}$: event frame, length-$k$ event context, and declared context domain for conditional laws and suprema.

H, $(\mathcal{N}, \mathscr{A}_{\mathcal{N}})$, $(\mathcal{Z}_H, \mathscr{A}_H)$, Z_{t+1}: prediction horizon, measurable mark space, complete measurable marked-time/no-event space, and observed outcome.

$a(x)$, $\mathscr{F}_t^{\mathrm{pred}}$, $\mathfrak{h}_t$, c_k, Replay_Θ, S_{Θ,t^-}: availability time, prediction information, observable history, context extractor, candidate replay operator, and reconstructed candidate state.

P_{obj}, P_{conf}, $P_\star$: design-generating law, confirmation-generating law, and externally fixed target law.

$\mathcal{S}_{\mathrm{obj}}$, $\mathcal{S}_{\mathrm{conf}}$: realized design sample or trajectory block and untouched confirmation sample or block.

$\nu(e)$, $\tau(e)$: event-mark extractor and declared scalar temporal anchor; the anchor is a point timestamp or, by default, interval onset.

$\mathcal{E}_\varnothing$, d_H: tagged no-event extension of the structured event space and fixed point-decision rule on forecast laws.

$\hat{e}_\theta^H(C)$, $\hat{e}_t^H(\mathbf{r})$, lift_H: coherent no-event-capable predictor summary, residual-record summary, and map aligning a structured template with a selected marked outcome.

Q_θ, $\mathcal{O}_\theta(C)$: predictive distribution over marked event times and the no-event outcome, and the typed bundle pairing that law with a coherent no-event-capable structured summary.

$\mathcal{L}_{\text{pred}}$, $\overline{\mathcal{L}}_{\text{pred}}$, $\mathcal{L}^{H}_{\text{event}}$: untransformed proper predictive loss, its preregistered bounded system-action transform, and bounded event-aware timing diagnostic.

$\mathcal{R}_{\text{pre}}$, $\mathcal{A}_{\text{post}}$: pre-observation admission risk and post-observation realized event action.

Q_B, B, b_t: baseline forecast law, conditional event-template predictor, and its template. Event buckets use K, never B.

$\mathscr{H}$, $\mathbb{H}^E_d$, $\mathbb{H}^Q_d$: finite-dimensional Hilbert space and separately tagged point-template and forecast-law copies of its self-adjoint operator representation space.

q_E, d_E, Π_E, δ_E, δ_Q: event encoder, decoder, admissibility projection, and point/law clipping radii.

$\oplus_E$: typed event residual composition $\mathcal{E} \times \mathbb{H}^E_d \to \mathcal{E}$.

$\mathcal{C}_{A,t^-}$, $\mathcal{C}_{R,t^-}$, $\mathcal{C}_E$: as-of exact-key residual, as-of general residual, and episodic caches.

$\mathcal{M}_R$, $\mathcal{V}_R$, $\mathbf{r}$, $\mathbf{0}_R$: residual-component mode set, typed record space, residual record, and no-correction record.

J^A_t, J^R_t, $\mathbf{r}^{\text{use}}_t$, $r^{E,\text{obs}}_{t,H}$: cache acceptance indicators, selected record, and point residual defined only for an in-horizon event.

ρ^Q_H, $r^{Q,\text{obs}}_{t,H}$, $\mathfrak{K}^Q_H$, Q^R_t: law-residual estimator, observed law residual, measurable full-outcome kernel map, and corrected law.

s_i, s_{k_t}, $\mathcal{S}_{\text{prov}}$: provenance records for general and exact residual-cache entries and their declared space.

Ξ_R, Ξ_B, $\Xi^{(v)}_A$, Λ_{eval}: residual contract, selective Bayesian contract, published versioned abstraction-compatibility structure, and the externally frozen evaluation contract.

H_i, H_{k_t}, v_{k_t}, v_t, μ_{k_t}: cache horizons, cache-entry and active epochs, and materialized compatibility safety margin.

$\mathcal{N}^B_t$, $\mathcal{R}^{\text{vec}}_t$, $\mathcal{N}^{\text{sh}}_t$: bounded Bayesian candidate frontier and its vector-retrieval and sheaf-inspired components.

A^B_t, τ^B_t, J^{act}_t: Bayesian activation score, criticality-adjusted threshold, and activation indicator.

$J^{\text{share}}_{K,t}$, q_{K,t^-}, $q^+_{K,t}$, $\mathcal{C}_{B,t^-}$: Anti-Pigeon posterior-sharing decision, cached prior, updated posterior, and as-of posterior cache.

$J^{\text{cp}}_{K,t}$, J^{audit}_t, U^{omit}_t: changepoint trigger, independent inactive-event audit indicator, and upper bound on local-versus-expanded forecast disagreement.

X_t, $\mathcal{X}_{\text{ctx}}$: compressed runtime state and its domain. This domain is distinct from the auxiliary event-field space $\mathcal{X}$.

$\mathcal{Y}_{\text{pkt}}$, $\mathcal{V}_Y$: runtime packet space and packet residual representation space.

B_Y, R_Y, $\oplus_Y$: packet baseline, packet residual, and typed packet composition.

$\widehat{\mathbf{y}}_{t+1}$, $\mathbf{y}^\star_{t+1}$: predicted and audited runtime packets.

$G_t = (V_t, R_t)$: time-unrolled typed event graph. Predictive-dependency and causal edges remain distinct.

$\mathfrak{M} = (U, V, F, P_U)$: structural causal model required for *do*-intervention notation.

$\pi : \mathcal{E} \to \mathcal{S}_{\mathrm{abs}}$, h_π, K, $\mathfrak{K}_\pi$, $\mathfrak{K}_\pi^+$: abstraction map, operational abstraction key including costed side information, one event bucket, all induced buckets, and active buckets with admissible contexts.

$\bar{e}_K$, $\mathfrak{C}_K$, $\mathcal{R}_C(K)$: concrete traceability frame, contexts anchored in a bucket, and coverage-aware context audit set.

$D_K^\star$, D_K^{mdl}, $D_K^{\mathrm{audit},\star}$, $\widehat{D}_K^\star$, $D_K^{\mathrm{cert},\star}$: external target-law future-diameter, model-only diagnostic diameter, restricted external diameter, its estimator, and simultaneous statistical-plus-continuity certificate.

$\overline{L}_K^{\mathrm{cert}}$: analytic uniform continuity bound or simultaneous upper confidence bound included in the bucket certificate.

D_Y^{law}: distance between probability laws. It is distinct from packet decoder d_Y.

$\mathcal{G}_t^A = (V_t^A, E_t^A)$: abstraction compatibility graph.

Q_i, $\mathcal{Y}_i$: node-local predictive law and its outcome space.

g_{ie}, r_{ie}: measurable node-to-edge comparison map and its pushforward restriction on predictive laws.

δ_e, Δ_{comp}: edge compatibility defect and maximum defect upper confidence bound.

$\mathfrak{Q}_{\mathcal{N}}$, D_i, $(\overline{\mathsf{Q}}_i)_{i \in \mathcal{N}}$: reconciliation feasible tuple family, local fidelity divergences, and returned reconciled forecast tuple.

∂_A, L_A: compatibility boundary and Laplacian, defined only for the stated finite-dimensional linear representation.

D_t^{design}, D_t^{conf}, $\mathfrak{S}_t(\Xi_A^{(v)}; \mathcal{N})$, Ψ_t, d_{edit}, $\mathcal{C}_{\mathrm{snap}}$: disjoint chronological snap-design and confirmation blocks, finite local candidate family, design-block score, structural-churn penalty, and revision cost. $\Theta_\Gamma[\Xi']$ is the complete candidate induced by an edit, including dependent local abstraction, model, key, and certificate revisions.

$\mathfrak{O}_t$, U_{obl}: externally fixed comparison-obligation set and weighted unresolved burden.

λ_{comp}, λ_{edit}, λ_{snap}, λ_{obl}: non-negative snap-selection weights for compatibility defect, structural churn, revision cost, and unresolved obligations; the final weight is strictly positive unless unresolved obligations are forbidden.

$\mathcal{D}_{\Delta,t}(\Xi')$, $\mathfrak{K}_{\Delta,t}(\Xi')$, $E_{\Delta,t}^{\mathrm{keep}}(\Xi')$: reverse dependency closure of a snap candidate and its affected active-bucket and retained/new-edge projections.

$\Delta\mathcal{R}_{\mathrm{prop},t}^{\mathrm{snap}}$, $G_{v \to \mathrm{cand},t}^{\mathrm{pri}}$, $C_{v \to \mathrm{cand},t}(h)$, A_t^{snap}: empirical confirmation proper-risk change, paired priority gain, utility-normalized resource cost, and joint snap-acceptance indicator.

T_{generate}, T_{obl}, T_{confirm}, T_{publish}: bounded snap-candidate generation, comparison-obligation validation, confirmation scoring, and atomic publication costs.

$\delta_{\mathrm{snap}} > 0$, $\epsilon_{\mathrm{obl}} \geq 0$, $\epsilon_{\mathrm{acc}}^{\mathrm{comp}} \in [0, \epsilon_{\mathrm{split}}^{\mathrm{comp}}]$: predeclared net-gain safety margin, unresolved-obligation limit, and compatibility threshold for affected retained or newly added edges.

$\mathbf{v}^{(v)}$, $\mathcal{C}_{\mathrm{mem}}^{(v)}$, $\mathsf{B}_{\mathcal{D}}$, $\mathsf{I}_{\mathcal{D}}$: local epoch map, versioned memory tuple, monotone epoch-bump map, and targeted stale-marking operator for a dependency closure.

$\mathcal{U}_0, \ldots, \mathcal{U}_5$, r_n, $d_t(h)$: baseline/refinement operators, selected stage at invocation n, and deepest reached stage under hardware profile h; $\mathcal{U}_3$ is bounded predictive sheaf snapping.

$\mathcal{B}_0, \ldots, \mathcal{B}_3$: bounded cached Bayesian update, bounded changepoint monitor, event-pattern refinement, and deep particle or variational inference stages.

p_t^{pri}, w_{pri}, $\mathcal{R}^D_{\mathrm{pri}}$, $\mathcal{R}^D_{\mathrm{prop}}$, $G^{\mathrm{pri}}_{a\to b}$: pre-outcome priority, its declared importance function, normalized weighted risk, unweighted proper risk, and gain between complete policies.

$\Delta_{\mathrm{pred}}(d)$, $\widehat{\Delta}_{\mathrm{pred}}(d)$, $s^{\mathrm{pred}}_{\mathrm{eff}}$, $s^{\mathrm{causal}}_{\mathrm{eff}}$: paired proper-risk effect of ablating distinction d, its confirmation estimate, the simultaneous-confidence-classified predictive sparsity ratio, and the separately identified causal sparsity ratio.

ζ_t, $\mathcal{Z}_{\mathrm{reg}}$: observed operating regime and its space. A regime is not causal by default.

$\mathcal{C}_{\mathrm{rep}}$, Φ: representation/runtime cost and finite-state descent potential.

$\mathfrak{F}^{\Gamma,\star}_{AP}$, $\mathcal{J}^{\mathrm{oracle}}_{\Gamma}$: population Anti-Pigeon-feasible design family and its oracle infimal benchmark.

$\mathfrak{G}_\Gamma$, $\widehat{\mathfrak{F}}^{\Gamma}_{AP}$, $\widehat{\Theta}_\Gamma$: finite predeclared design family, empirically certified feasible family, and operationally selected design.

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